

THE UNIVERSITY OF NEW SOUTH WALES
SCHOOL OF PHYSICS
THIRD YEAR LABORATORY

CHAOTIC MOTION

CM ()

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Chaotic Motion

1. **Excel for plotting** “Position vs Time” (Expts 1 and 2)
2. **Excel for plotting** “Poincare plots” (Expts 3 and 4)
3. **Book extract:** Baker & Gollub “Chaotic Dynamics”

Chaotic Motion

“Excel” instructions for plotting position vs time (Expts 1, 2)

Open “Excel”

File

Open...

Look in:

My Documents

Daedalon

open your data folder

Files of type: choose ‘All Files’

open your PHA data file’

Open

Text import wizard -

Step 1 of 3 Original data type - choose ‘Delimited’
Start import at row: 12
(File origin: 437 : OEM United States)
Next >

Step 2 of 3 Delimiters - tick ‘Space’ (leave ‘Tab’, leave ‘Treat
consecutive delimiters as one’)

Finish

Now left click A (at the top of column A) to highlight column A, and drag to B to highlight both columns A and B

Insert

Chart...

Chart wizard –

Step 1 of 4 – Chart Type

Standard Types

Chart type: XY (Scatter)

Chart sub-type: bottom option on the right
(Scatter with data points connected
by lines without markers)

Next >

Next >

Step 3 of 4 - Chart Options,

Titles: enter Chart title: (e.g. “16 mm damping”)

(Axes: leave both Value (X) and (Y) axis ticked)

Gridlines: tick Major gridlines for both (X) and (Y) axis

Legend: **untick** ‘Show legend’

(Data labels: leave all unticked)

Finish

Hover over the gray graph area and **right** click 'Plot Area'
then select 'Clear', to delete the grey colour

Hover over the X axis and right click 'Value (X) axis'
then select 'Format Axis...'

click the 'Scale' tab and enter 'Value (X) axis scale'
enter desired Minimum and Maximum values

OK

Hover over the Y axis and right click 'Value (Y) axis'
then select 'Format Axis...'

click the 'Scale' tab and enter 'Value (Y) axis scale'
enter desired Minimum and Maximum values

OK

Left click anywhere on the chart, then

File

Print...

Chaotic Motion

“Excel” instructions for Poincaré plots (Expts 3, 4)

Open “Excel”

File

Open...

Look in:

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open your data folder

Files of type: choose ‘All Files’

open your POI data file’

Open

Text import wizard -

Step 1 of 3 Original data type - choose ‘Delimited’
Start import at row: 4
(File origin: 437 : OEM United States)
Next >

Step 2 of 3 Delimiters - tick ‘Space’ (leave ‘Tab’, leave ‘Treat
consecutive delimiters as one’)

Finish

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Insert

Chart...

Chart wizard –

Step 1 of 4 – Chart Type

Standard Types

Chart type: XY (Scatter)

Chart sub-type: top option (Scatter, Compares
pairs of values)

Next >

Next >

Step 3 of 4 - Chart Options,

Titles: enter Chart title: (e.g. “phase = 0.8”)

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Gridlines: tick Major gridlines for both (X) and (Y) axis

Legend: **untick** ‘Show legend’

(Data labels: leave all unticked)

Finish

Hover over the gray graph area and **right** click 'Plot Area'
then select 'Clear', to delete the grey colour

Hover over the X axis and right click 'Value (X) axis'
then select 'Format Axis...'

click the 'Scale' tab and enter 'Value (X) axis scale'
enter desired Minimum and Maximum values

OK

Hover over the Y axis and right click 'Value (Y) axis'
then select 'Format Axis...'

click the 'Scale' tab and enter 'Value (Y) axis scale'
enter desired Minimum and Maximum values

OK

Left click anywhere on the chart, then

File

Print...

Chaotic dynamics

an introduction

SECOND EDITION

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Bryn Athyn, Pennsylvania*

AND

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Introduction

The irregular and unpredictable time evolution of many nonlinear systems has been dubbed 'chaos.' It occurs in mechanical oscillators such as pendula or vibrating objects, in rotating or heated fluids, in laser cavities, and in some chemical reactions. Its central characteristic is that the system does not repeat its past behavior (even approximately). Periodic and chaotic behavior are contrasted in Figure 1.1. Yet, despite their lack of regularity, chaotic dynamical systems follow deterministic equations such as those derived from Newton's second law.

The unique character of chaotic dynamics may be seen most clearly by imagining the system to be started twice, but from slightly different initial conditions. We can think of this small initial difference as resulting from measurement error, for example. For nonchaotic systems this uncertainty leads only to an error in prediction that grows *linearly* with time. For chaotic systems, on the other hand, the error grows *exponentially* in time, so that the state of the system is essentially unknown after a very short time. This phenomenon, which occurs only when the governing equations are nonlinear, is known as *sensitivity to initial conditions*. Henri Poincaré (1854–1912), a prominent mathematician and theoretical astronomer who studied dynamical systems, was the first to recognize this phenomenon. He described it as follows: '...it may happen that small differences in the initial conditions produce very great ones in the final phenomena. A small error in the former will produce an enormous error in the latter. Prediction becomes impossible, and we have the fortuitous phenomenon' (Poincaré, 1913).

If prediction becomes impossible, it is evident that a chaotic system can resemble a stochastic system (a system subject to random external forces). However, the source of the irregularity is quite different. For

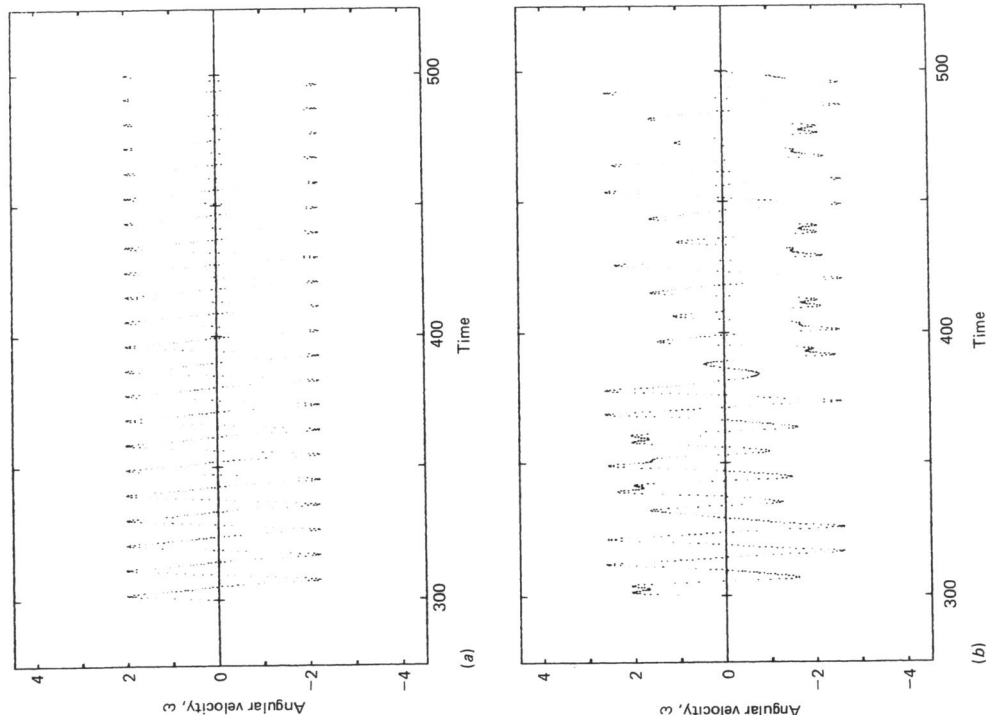


Fig. 1.1 The damped, driven pendulum can exhibit both periodic and chaotic motions. Here, the angular velocity is shown as a function of time for the two cases.

chaos, the irregularity is part of the intrinsic dynamics of the system, not unpredictable outside influences.

Chaotic motion is not a rare phenomenon. Consider a dynamical system described by a set of first order differential equations. Necessary conditions for chaotic motion are that (a) the system has at least three independent dynamical variables, and (b) the equations of motion contain a nonlinear term that couples several of the variables. The equations can often be expressed in the form:

$$\left. \begin{aligned} dx_1/dt &= F_1(x_1, x_2, \dots, x_n), \\ dx_2/dt &= F_2(x_1, x_2, \dots, x_n), \\ &\vdots \\ dx_n/dt &= F_n(x_1, x_2, \dots, x_n), \end{aligned} \right\} \quad (1.1)$$

where n must be at least 3. Two examples of appropriate nonlinear equations are:

$$\left. \begin{aligned} dx_1/dt &= \alpha x_1 + \beta x_2 + \gamma x_1 x_2 + \dots + \delta x_n, \\ dx_2/dt &= \alpha x_1 + \beta x_2 + \gamma \sin x_2 + \dots + \delta x_n \end{aligned} \right\} \quad (1.2)$$

where $\alpha, \beta, \gamma, \delta$ are constants. In each case the nonlinear term couples both x_1 and x_2 . Systems such as these are often chaotic for some choices of the constants.

The fact that only three variables are required for chaos was surprising when first discovered. We shall see that three-space is sufficient to allow for (a) divergence of trajectories, (b) confinement of the motion to a finite region of the phase space of the dynamical variables, and (c) uniqueness of the trajectory. The nonlinearity condition is perhaps less surprising. Solutions to linear differential equations can always be expressed as a linear superposition of periodic functions, once initial transients have decayed. The effect of a nonlinear term is often to render a periodic solution unstable for certain parameter choices. While these conditions do not guarantee chaos, they do make its existence possible.

The nonlinearity condition has probably been responsible for the late historical development of the study of chaotic systems. Despite the fact that chaotic systems are deterministic and are described by many of the long-known classical equations of physics, the development of the subject itself is more recent. This circumstance may arise from the fact that, with the exception of some first order equations, nonlinear differential equations are either difficult or impossible to solve analytically. Although it is sometimes possible to use linearized approximations, the solution of nonlinear differential equations generally requires numerical methods whose practical implementation demands the use of a digital computer. The first numerical study to detect chaos in a nonlinear dynamical system was that of Lorenz's model of convective fluid flow (Lorenz, 1963). Similarly, the majority of the diagrams in this book are based upon the use of numerical methods on a personal computer to solve nonlinear equations.

From these general comments on chaotic systems, we turn to the physical system that is the focus of this work – the damped, driven pendulum. The choice of the pendulum as a model system has strong

historical precedent in physics. Galileo postulated the constancy of period for small amplitude oscillations of the pendulum from observations of swaying lamps in the cathedral at Pisa in 1581 (Robinson, 1921). He took up the problem of the relationship between the period and pendulum length in his famous *Dialogue on the Two Principal World Systems* in 1632, and in 1637 he suggested that the square of the period was proportional to the length of the pendulum for small oscillation amplitudes (Dugas, 1958). The pendulum also served as a primary timing mechanism for clocks and as a method of measuring variations in the earth's gravitational field. As a pedagogical device the pendulum has long been a standard mechanical example in introductory physics and classical mechanics courses. Now, 400 years after Galileo's initial work, the pendulum has again become an object of research as a chaotic system. The references scattered throughout this work attest to its popularity.

The damped, sinusoidally driven pendulum of mass m (or weight W) and length l is described by the following equation of motion:

$$ml^2 \frac{d^2\theta}{dt^2} + \gamma \frac{d\theta}{dt} + Wl \sin \theta = A \cos(\omega_D t). \quad (1.3)$$

This equation expresses Newton's second law with the various terms on the left representing acceleration, damping, and gravitation. The angular velocity of the forcing, ω_D , may be different from the natural frequency of the pendulum. In order to minimize the number of adjustable parameters the equation may be rewritten in dimensionless form as:

$$d^2\theta/dt^2 + (1/q)d\theta/dt + \sin\theta = g\cos(\omega_D t) \quad (1.4)$$

where q is the damping or quality parameter, g is the forcing amplitude, *not to be confused with the gravitational acceleration*, and ω_D is the drive frequency. The low-amplitude natural angular frequency of the pendulum is unity, and time is regarded as dimensionless. (This particular notation follows that used by Gwinn and Westervelt. See, for example, Gwinn and Westervelt (1986).) This equation satisfies the necessary conditions for chaos when it is written as a set of first order equations:

$$\left. \begin{aligned} d\omega/dt &= -(1/q)\omega - \sin\theta + g\cos\phi, \\ d\theta/dt &= \omega, \\ d\phi/dt &= \omega_D. \end{aligned} \right\} \quad (1.5)$$

The variable ϕ is introduced as the phase of the drive term. The

necessary three variables (ω, θ, ϕ) are evident, and the $\sin\theta$ and $g\cos\phi$ terms are clearly nonlinear. Whether the motion is chaotic depends upon the values of the parameters g , ω_D , and q . For some values the pendulum locks onto the driving force, oscillating in a periodic motion whose frequency is the driving frequency, possibly with some harmonics or subharmonics. But for other choices of the parameters the pendulum motion is chaotic. One may view the chaos as resulting from a subtle interplay between the tendency of the pendulum to oscillate at its 'natural' frequency and the action of the forcing term. The transitions between nonchaotic and chaotic states, due to changes in the parameters, occur in several ways and depend delicately upon the values of the parameters.

A variety of analytic and computational tools may be used in the study of chaotic systems. In Chapter 2 several of these are discussed. The pendulum's phase space and its properties are described, together with the conceptual device known as the Poincaré section. Then, since Fourier spectra are an indicator of chaotic motion, some elements of Fourier analysis are outlined. Chapter 3 is a description of the application of these and other techniques to the pendulum.

The driven pendulum would seem to be one of the simplest physical systems. Yet its behavior is rich and complex. The study of its motion can be facilitated by simple mathematical models formulated as difference equations, that provide a discrete *mapping* of the system from one state to another. Mappings have the advantage of being conceptually simple and numerically efficient, and they may be used as paradigms for various aspects of the pendulum motion. Chapter 4 contains discussions of three such maps, the logistic map, the circle map, and the horseshoe map. We use them to provide insight into the behavior of the pendulum.

Chapter 5 is concerned with the geometric structure of the *attractor* that describes the chaotic pendulum. The attractor, and its Poincaré section, are *fractal* structures with noninteger dimensionality. Various approaches to the calculation of fractal dimension are described. Another geometric feature is the exponential divergence of the chaotic trajectories on the attractor. The rate of this divergence is characterized by Lyapunov exponents. The calculation of these exponents and their relation to (a) the fractal dimension, (b) the dissipative nature of the pendulum, and (c) the duration of predictable behavior are also discussed.

Up to this point the presentation is focused on the fundamental ideas of chaotic dynamics. In Chapter 6 we discuss the relationship between these ideas and the analysis of experimental data. The developing methodology for characterization of nonlinear dynamical behavior in

experimental phenomena is complex. In this chapter we describe some of these methods and apply them to experimental data from a physical pendulum. The results of this study are then compared to those from the numerical simulations developed earlier in the book. The experimental data are also used to illustrate the possibility of prediction of chaotic data. Finally, we illustrate, through numerical simulation, the control of unstable dynamical states, in an otherwise chaotic pendulum.

Chapter 7 concludes the book with a brief survey of chaotic behavior in physical systems, including lasers, chemical reactions, fluids, crystal growth, and earthquakes. We emphasize the extension of chaotic dynamics to spatially extended systems having many degrees of freedom. Finally, the application of chaotic dynamics to quantum systems, and the connection between chaos and irreversibility are also discussed briefly.

Two appendices present numerical aspects of this book. Appendix A is a description of the Runge–Kutta algorithm used to solve the pendulum differential equation. Appendix B provides brief descriptions and listings of the computer programs used throughout the text, and in the computer exercises given at the end of several of the chapters. The listings utilize the language True BASICtm, but they are adaptable to any compiled BASIC or other high level language. (Interpreted BASIC, which is typically delivered with current microcomputers, is too slow for most of these simulations. The exceptions are the mappings in Chapter 4.) A third appendix, C, provides solutions to selected problems.

2 Some helpful tools

In this chapter we discuss three mathematical constructs that are generally useful in the study of dynamical systems: phase space, the Poincaré section, and power spectra. Phase space is the mathematical space of the dynamical variables of a system. The Poincaré section is a ‘snapshot’ of the motion in the phase space, taken at regular time intervals. The power spectrum is computed using Fourier analysis to display the frequency composition of the time variation of the dynamical variables.

2.1 Phase space

The phase space of a dynamical system is a mathematical space with orthogonal coordinate directions representing each of the variables needed to specify the instantaneous state of the system. For example, the state of a particle moving in one dimension is specified by its position (x) and velocity (v); hence its phase space is a plane. On the other hand, a particle moving in three dimensions would have a six-dimensional phase space with three position and three velocity directions. A phase space may be constructed in several different ways. For example, momenta can be used instead of velocities.

Let us focus the discussion on the pendulum and begin with the familiar simple pendulum in the small amplitude approximation where the restoring term, $\sin\theta$, is taken as θ . (Recall that the equations are written in dimensionless form for simplicity, with time measured in units of the inverse of the natural frequency.) The equation of motion is

$$d^2\theta/dt^2 + \theta = 0. \quad (2.1)$$

With the addition of the angular velocity variable, $\omega \equiv d\theta/dt$, this linear, second order equation can be reduced to two first order equations:

$$d\omega/dt = -\theta$$

and

$$d\theta/dt = \omega.$$

In this way each dynamical variable has its own first order differential equation. Without loss of generality, the initial conditions can be chosen so that the solution becomes

$$\theta = a_1 \cos t \text{ and } \omega = a_1 \sin t$$

where $\{a_i\}$ represents the possible amplitudes of the motion. This solution set gives the parametric curves for ω and θ , and one can eliminate the time parameter to give a two-dimensional representation for differing values of a_i . This diagram, shown in Figure 2.1, is the appropriate *phase space diagram* (in this case a phase plane diagram). Each value of a_i yields a closed orbit of fixed energy. The energy increases with the square of the radius a_i . The orbit is usually called a *phase trajectory*.

An important feature of the trajectory is that two trajectories corresponding to similar energies will pass very close to each other, but the orbits will not cross each other. This *noncrossing* property derives from the fact that past and future states of a deterministic mechanical

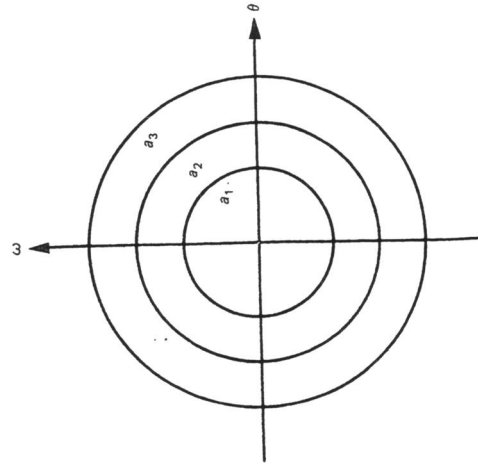


Fig. 2.1 Phase diagram of the linear pendulum. The angular velocity, ω , and the angular displacement, θ , are the coordinate axes.

system are uniquely prescribed by the system state at a given time. A crossing of trajectories at time t would introduce ambiguity into past and future states, thereby rendering the system indeterminate. Such indeterminacy would contradict the assumed uniqueness of the trajectory. Figure 2.2 shows the indeterminacy of trajectories emanating from a hypothetical crossing.

Another important feature of the phase space of *conservative* (constant energy) systems is the *preservation of areas*. This means that all the points found in a given area of phase space at one time move in such a way that at a later time the area occupied by these points remains the same. This feature is illustrated in Figure 2.3 and in Examples 2.1 and 2.2.

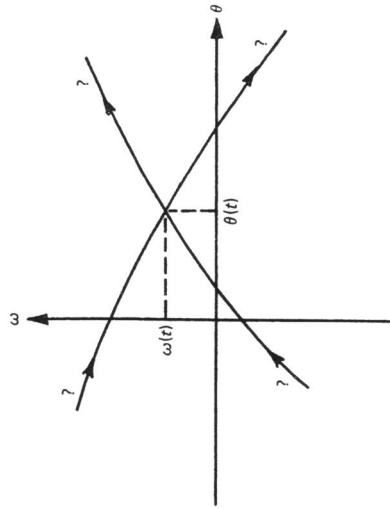


Fig. 2.2 The noncrossing property of phase trajectories. Crossing of trajectories violates uniqueness of trajectories in a deterministic dynamical system.

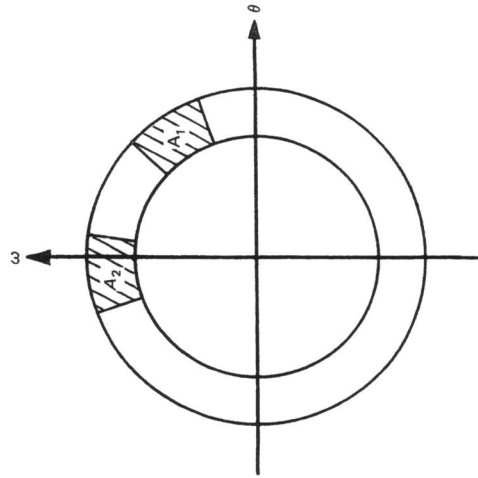


Fig. 2.3 Preservation of phase space area.

Example 2.1

For the linear oscillator $d^2\theta/dt^2 + \theta = 0$, consider the evolution of the area A_1 as shown in Figure 2.4 during one quarter of the period. Since the system is energy conserving, A_1 should remain constant. Because of the circular symmetry, preservation of the area can be shown by proving that every point in A_1 rotates (at a constant radius) through the same angle in the quarter period. The energy conserving feature ensures that each point rotates at a constant radius because the energy of the oscillator is proportional to the square of the radius. For the rotation angle we note that since $\theta = a \cos t$ and $\omega = a \sin t$, the polar angle of a given point is

$$\alpha(t) = \tan^{-1}(\tan t) = t. \quad (2.2)$$

Therefore at $t = t_0 + \pi/2$, $\alpha(t_0 + \pi/2) = t_0 + \pi/2$. But since $\alpha(t)$ was arbitrary, all points rotate by $\pi/2$ in one quarter period, and the area is preserved.

Example 2.2

As another example of area preservation consider the very simple motion of a constant velocity rotor. The two first order equations become

$$d\omega/dt = 0$$

and

$$d\theta/dt = \omega_0.$$

The corresponding phase trajectories are just horizontal lines with differing angular velocities, ω_{0i} , as shown in Figure 2.5. The linear dependence of θ on ω_{0i} ensures that an initial rectangle of points transforms to a parallelogram with a constant base and height, thereby maintaining the original area.

Example 2.2 also raises the question of boundaries on the phase plane coordinates. In contrast to the linearized pendulum whose finite motion allowed both θ and ω to be bounded conveniently in phase space, the angular coordinate θ for the rotor can increase (positively or negatively) without bound. Yet physically θ is periodic. Therefore the phase

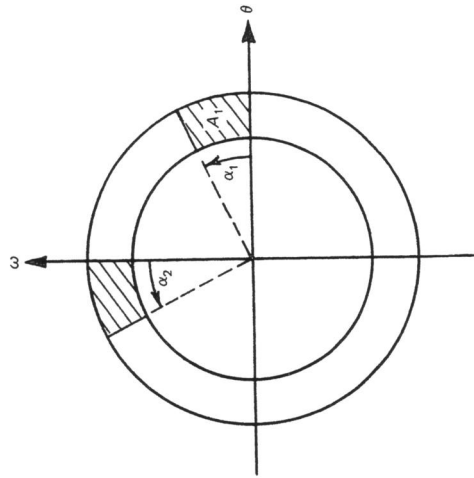


Fig. 2.4 Evolution of the linear oscillator described in Example 2.1.

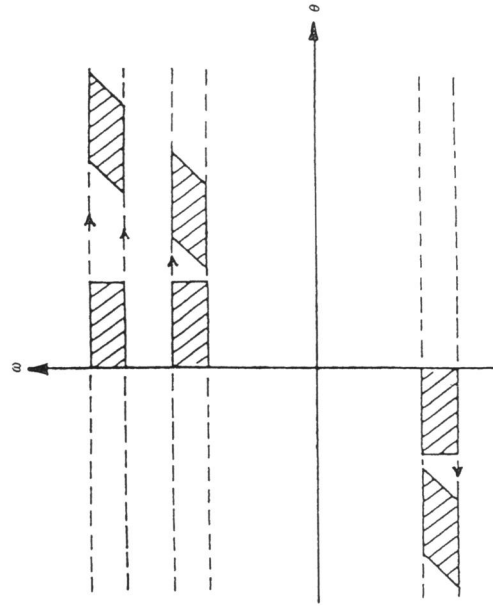


Fig. 2.5 Phase space diagram of a constant velocity rotor.

diagram is also made periodic by imposing *periodic boundary conditions* on θ as illustrated in Figure 2.6. The θ axis can be limited to $[-\pi, \pi]$, and the two edges of this domain are regarded as identical. As the rotor goes around in the positive θ direction, its phase representation disappears off the right edge of the phase diagram and immediately reappears on the left side. Similar periodic boundary conditions can also be usefully

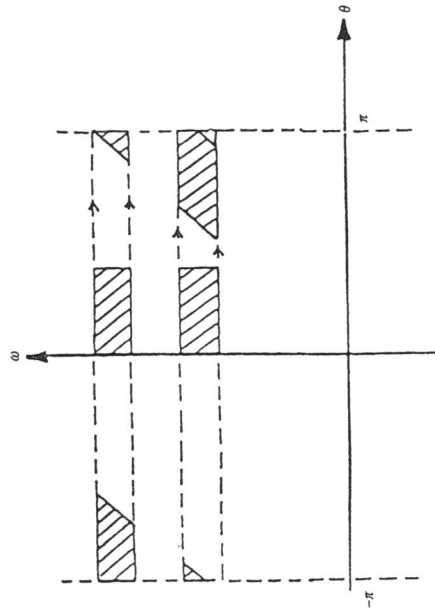


Fig. 2.6 Phase space diagram of the rotor with periodic boundary conditions. Phase points moving to the right disappear at $\theta = \pi$ and then reappear at $\theta = -\pi$.

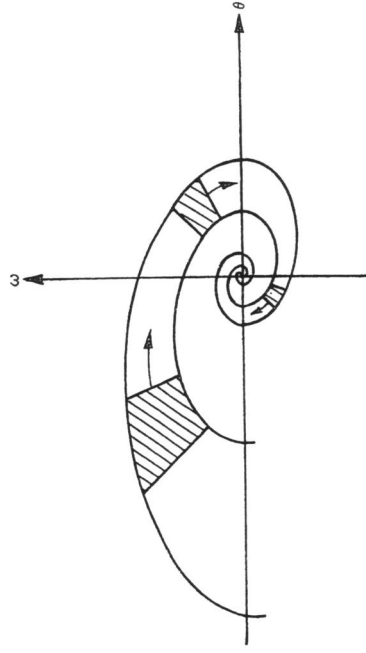


Fig. 2.7 Phase space diagram of the dissipative linear pendulum. Phase area is not preserved.

applied to the forced pendulum whose motion passes through the vertical direction.

The property of area preservation, or volume preservation in a higher-dimensional space, is a general feature of conservative systems. This property leads to a classification of dynamical systems into two categories – *conservative* or *dissipative* – depending upon whether the phase volumes stay constant or contract, respectively. For example, the linearized undamped pendulum conserves energy, and its trajectories preserve phase area. On the other hand, the trajectories of the linearized damped pendulum,

$$d^2\theta/dt^2 + d\theta/dt + \theta = 0, \quad (2.3)$$

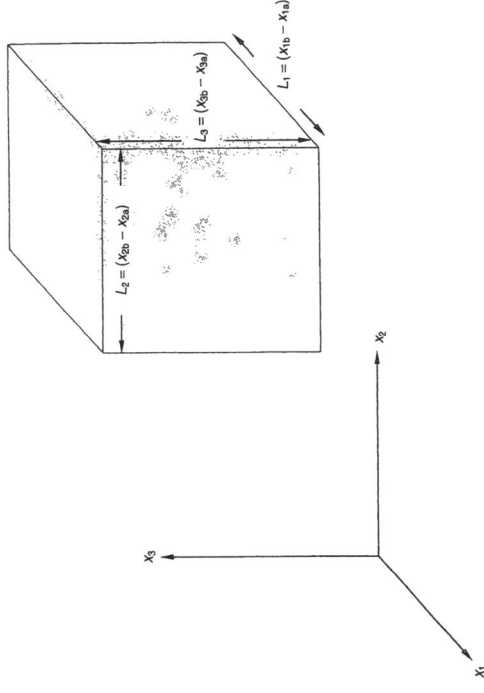


Fig. 2.8 Evolution of a volume in three-dimensional phase space.

decay to a single point: $\omega = 0 = \theta$. This area contraction is illustrated in Figure 2.7. Such a point is called an *attractor*, because a finite set of initial coordinates (θ, ω) converge to it. Obviously phase area is *not* preserved and the system is said to be dissipative.

Using these phase space characteristics we can develop a method for determining from the equations of motion whether a system is conservative or dissipative. The development of the method is easiest to understand in three space dimensions; thus we assume phase coordinates x_1, x_2, x_3 , as in Figure 2.8. The equations of motion of the system can be written in terms of the phase ‘velocity’ components,

$$\begin{cases} dx_1/dt = F_1(x_1, x_2, x_3), \\ dx_2/dt = F_2(x_1, x_2, x_3), \\ dx_3/dt = F_3(x_1, x_2, x_3). \end{cases} \quad (2.4)$$

Now consider a set of phase points contained in a small ‘box’ in phase space of volume V with sides of lengths L_1, L_2 , and L_3 , as shown in Figure 2.8. In a short time dt this box will evolve to a slightly different one with volume $V + dV$. The length L_j is given by the difference in phase coordinates, $(x_{jb} - x_{ja})$. All lengths are small quantities.

The volume of the ‘box’ is given by

$$\begin{aligned} V &= L_1 L_2 L_3 \\ &= (x_{1b} - x_{1a})(x_{2b} - x_{2a})(x_{3b} - x_{3a}). \end{aligned}$$

Then, using the product rule for differentiation, the rate of change of volume becomes

$$dV/dt = L_2 L_3 (\dot{x}_{1b} - \dot{x}_{1a}) + L_1 L_3 (\dot{x}_{2b} - \dot{x}_{2a}) + L_1 L_2 (\dot{x}_{3b} - \dot{x}_{3a}), \quad (2.5)$$

where $\dot{x}$ indicates dx/dt . This expression can be rewritten, using equation (2.4), as

$$dV/dt = L_2 L_3 [F_1(x_{1b}, x_{2a}, x_{3a}) - F_1(x_{1a}, x_{2a}, x_{3a})] + 2 \text{ similar terms.}$$

Next we approximate with a Taylor series, for example,

$$F_1(x_{1b}, x_{2a}, x_{3a}) \approx F_1(x_{1a}, x_{2a}, x_{3a}) + \frac{\partial F_1(x_{1a}, x_{2a}, x_{3a})}{\partial x_1} (x_{1b} - x_{1a}). \quad (2.6)$$

Then the rate of change of volume is

$$\begin{aligned} \frac{dV}{dt} &= L_2 L_3 [F_1(x_{1a}, x_{2a}, x_{3a}) + \frac{\partial F_1}{\partial x_1} (x_{1b} - x_{1a}) \\ &\quad - F_1(x_{1a}, x_{2a}, x_{3a})] + 2 \text{ similar terms} \\ &= L_2 L_3 \frac{\partial F_1}{\partial x_1} (x_{1b} - x_{1a}) + 2 \text{ similar terms.} \end{aligned}$$

Finally, the logarithmic rate of volume change is compactly expressed as

$$\frac{1}{V} \frac{dV}{dt} = \mathbf{V} \cdot \mathbf{F}. \quad (2.7)$$

The logarithmic derivative is therefore independent of the particular volume chosen, and depends only on $\mathbf{V} \cdot \mathbf{F}$. If this quantity is zero, the system is termed *conservative*, whereas if the divergence of phase velocity is negative the system is *dissipative*. The kinematic properties of the flux in phase space for a conservative system are analogous to the flow of an incompressible fluid in hydrodynamics. (The term *Hamiltonian* is sometimes used in connection with phase volume preserving systems. Many dynamical systems obey Hamilton's equations of motion and such systems are called *Hamiltonian* systems. These systems preserve volume in phase space, according to Liouville's theorem, and therefore Hamiltonian systems are a subset of the set of conservative systems. See Helleman (1983).)

Let us write both of our example pendula in the phase velocity form and determine their phase space preservation characteristics by this method

(i) $d^2\theta/dt^2 + \theta = 0$ (undamped) becomes

$$d\theta/dt = \omega \text{ and } d\omega/dt = -\theta.$$

Therefore $\mathbf{F} = (\omega, -\theta)$ and $\mathbf{V} \cdot \mathbf{F} = \partial\omega/\partial\theta + \partial(-\theta)/\partial\omega = 0$, indicating that phase area is preserved.

(ii) $d^2\theta/dt^2 + d\theta/dt + \theta = 0$ (damped) becomes

$$d\theta/dt = \omega \text{ and } d\omega/dt = -\omega - \theta.$$

Therefore $\mathbf{F} = (\omega, -\omega - \theta)$ and $\mathbf{V} \cdot \mathbf{F} = \partial\omega/\partial\theta + \partial(-\omega - \theta)/\partial\omega = -1$, indicating that phase area diminishes in time and the system is, as expected, dissipative.

These two examples show how easily the divergence criterion may be applied. For the driven pendulum, the equation:

$$d^2\theta/dt^2 + (d\theta/dt)/q + \sin\theta = g\cos(\omega_D t)$$

is converted to a set of first order equations:

$$d\omega/dt = -\omega/q - \sin\theta + g\cos\phi,$$

$$d\theta/dt = \omega,$$

$$d\phi/dt = \omega_D.$$

Then the right sides form the components of the three-dimensional vector $\mathbf{F}$. It is left as an exercise for the reader to show that $\mathbf{V} \cdot \mathbf{F} = -1/q$ and that, therefore, the system is dissipative.

We have looked at phase diagrams for the damped and undamped linearized pendula. Let us now introduce the full nonlinear restoring torque, $\sin\theta$. Figure 2.9 shows the phase plane for the undamped pendulum:

$$d^2\theta/dt^2 + \sin\theta = 0. \quad (2.8)$$

For small values of $d\theta/dt$ and θ , the diagram appears similar to that of the pendulum in the linear approximation, but as θ approaches $\pm\pi - a$ the pendulum swing that would go all around the circle - the picture changes. At $(\pm\pi, 0)$ the slope develops a discontinuity. The largest of the closed trajectories bounds the region where the motion is oscillatory (or vibrational). On the open trajectory of higher angular velocity, the

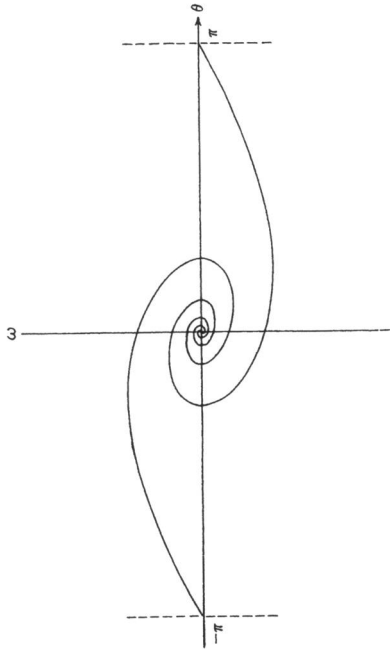


Fig. 2.11 Pair of phase space trajectories for the damped pendulum.

be seen by setting the phase velocity equal to zero and solving for the stationary values of θ and ω ; that is,

$$\left. \begin{aligned} d\theta/dt &= \omega = 0, \\ d\omega/dt &= -\omega - \sin\theta = 0. \end{aligned} \right\} \quad (2.10)$$

While these attractors are points where the phase velocity goes to zero, questions arise as to the stability of these points. Will the trajectories tend to go back to these critical points if slightly perturbed? Will the stability depend upon the direction of the perturbation? These questions can be answered by looking carefully at the critical points. A useful technique for examining dynamical behavior near critical points involves the assumption that the system will not deviate substantially from linear behavior *near the critical points*. Then each of the nonlinear terms in the differential equations is given a linear approximation near the critical points. This method was developed by Poincaré in 1914 (Hayashi, 1964).

For the case of the damped pendulum,

$$\left. \begin{aligned} d\theta/dt &= \omega, \\ d\omega/dt &= -\omega - \sin\theta, \end{aligned} \right\} \quad (2.11)$$

it is easy to see (Problem 2.6) that near $\theta = \pm n\pi$, where n is even, the linear approximation is

$$\left. \begin{aligned} d\theta/dt &= \omega, \\ d\omega/dt &= -\omega - (\theta - n\pi), \end{aligned} \right\} \quad (2.12)$$

and when n is odd the linear approximation becomes

$$\left. \begin{aligned} d\theta/dt &= \omega, \\ d\omega/dt &= -\omega + (\theta - n\pi). \end{aligned} \right\} \quad (2.13)$$

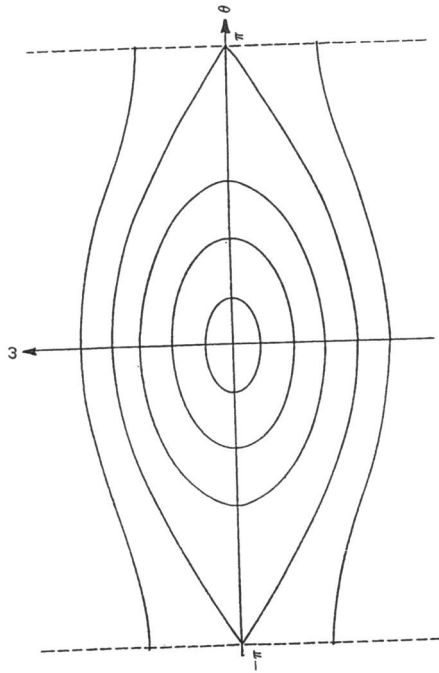


Fig. 2.9 Phase diagram of the nonlinear pendulum. The restoring force term contains $\sin\theta$.

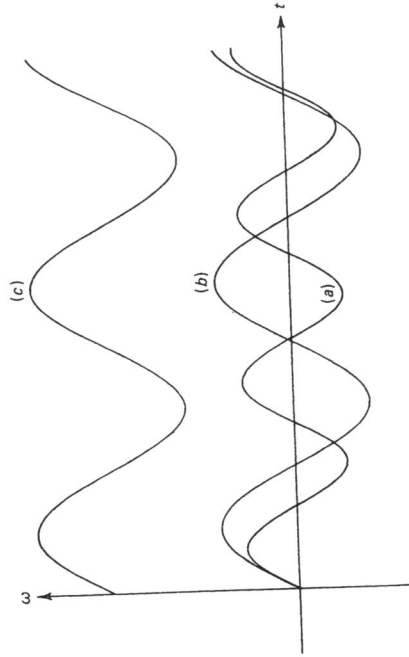


Fig. 2.10 Angular velocity time series. In contrast to the linearized pendulum, the period of the motion for the nonlinear pendulum increases with increasing amplitude. Curves (a) and (b) show oscillatory motions of differing amplitudes. Curve (c) shows the pendulum with sufficient energy to exhibit both rotary and oscillatory motions.

pendulum goes completely around the circle, and its motion is a rotation modulated by oscillation. The average angular velocity becomes nonzero. (One might compare this to a direct current electrical signal modulated by an alternating current signal.) The corresponding time series of these motions are shown in Figure 2.10.

Consider now the addition of a damping term so that the equation becomes

$$d^2\theta/dt^2 + d\theta/dt + \sin\theta = 0. \quad (2.9)$$

Typical trajectories are shown in Figure 2.11. As indicated previously, the damping term results in an attractor at the origin where $\sin\theta \approx \theta$. Now, however, further attractors are added at $\theta = \pm n\pi$, $\omega = 0$. This can

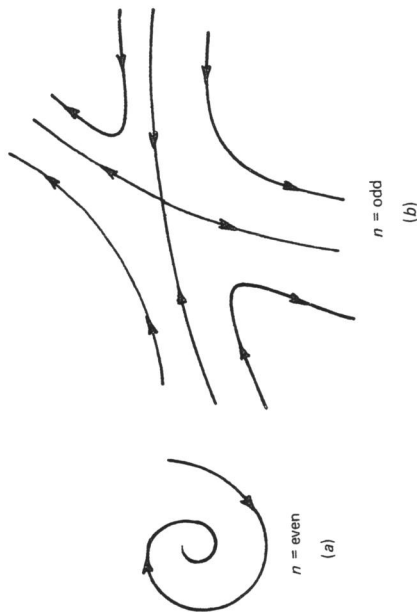


Fig. 2.12 Critical points in phase space: (a) focal point; (b) saddle point. In (b) the trajectories are stable whereas the trajectories coming from the saddle point are unstable.

In each case θ is transformed to a value centered at the critical point such that $\theta \rightarrow \Delta\theta = \theta - n\pi$ and therefore the linearized phase plane equations are

$$\left. \begin{aligned} n = \text{even}; \quad d\Delta\theta/dt = \omega, \quad d\omega/dt = -\omega - \Delta\theta, \\ n = \text{odd}; \quad d\Delta\theta/dt = \omega, \quad d\omega/dt = -\omega + \Delta\theta. \end{aligned} \right\} \quad (2.14)$$

Following the usual method for solving sets of first order linear differential equations, trial solutions of the form $\Delta\theta = Ae^{\lambda t}$ and $\omega = Be^{\lambda t}$ may be substituted into the equations; this yields two pairs of homogeneous algebraic equations. The condition for a nontrivial solution is the vanishing of the determinant of the coefficients of A and B . This condition produces quadratic characteristic equations in λ for each case:

$$\lambda^2 + \lambda + 1 = 0: n = \text{even}, \quad \lambda^2 + \lambda - 1 = 0: n = \text{odd}. \quad (2.15)$$

For the $n = \text{even}$ case, the λ values are complex conjugates with negative real parts. This implies that both $\Delta\theta$ and ω will spiral inward toward the equilibrium point attractor, which is called a *focus*. This case is shown in Figure 2.12(a). On the other hand, the $n = \text{odd}$ condition produces two real values of λ , one positive and one negative. In this case the stable phase trajectories move toward the critical point in one direction (negative exponent), but the unstable trajectories move away from it in another direction (positive exponent). This kind of critical point is called a *saddle point* and is shown in Figure 2.12(b). The respective directions are obtained by determining the A and B coefficients appropriate to each λ value. Note how the unstable trajectory directions correspond to the only possible directions for the pendulum, located momentarily at the saddle point. The details of the solution are left as an exercise (Problem

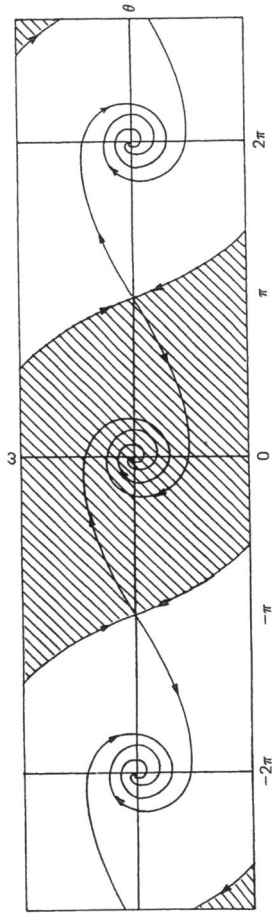


Fig. 2.13 Phase space diagram of the damped pendulum. Alternate shaded and unshaded regions are basins of attraction. All points within a particular basin are attracted to the focal point within the basin.

2.7). Putting all this information together, the phase diagram for the damped pendulum can be drawn as in Figure 2.13.

This phase diagram suggests yet another property of trajectories in phase space. As drawn in Figure 2.13, the phase space is divided into alternating regions as indicated by the shading. Inside each region all the trajectories will eventually spiral to the enclosed focal point. Each region is the set of all initial conditions (ω, θ) of trajectories that will eventually converge on a specific attractor – in this case a focal point. Such regions are called *basins of attraction*. Furthermore, each of the diagonal curves (Figure 2.13) dividing one basin from another is called a *separatrix*. The arrows on the separatrix (and elsewhere) indicate the flow of the trajectories, toward and away from the saddle points. We will see (Chapter 3) that one characteristic of chaos is the partial dissolution of the separatrix as the basins start to merge.

While most of the discussion so far has focused on the phase plane, it is important to realize that the phase space construction need not be confined to two dimensions. A previously introduced equation set,

$$\left. \begin{aligned} dx_1/dt &= F_1(x_1, x_2, x_3), \\ dx_2/dt &= F_2(x_1, x_2, x_3), \\ dx_3/dt &= F_3(x_1, x_2, x_3), \end{aligned} \right\} \quad (2.16)$$

would define a three-dimensional phase space. We may use this set of equations to illustrate some further properties of phase space. We have already described the divergence method for determining if the above system is conservative. Further, it is evident that the equations are not explicitly time-dependent. The equation set is then called *autonomous*

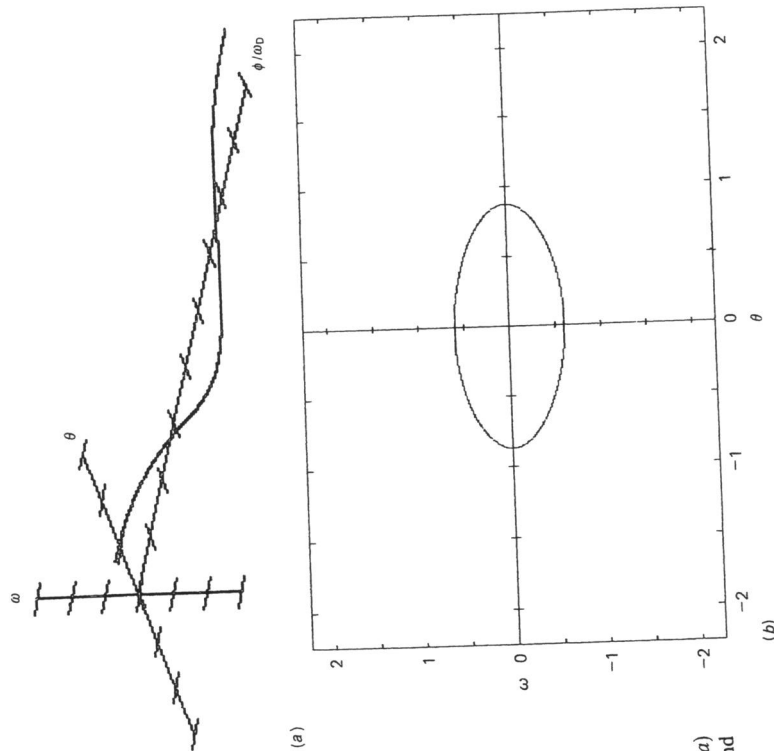


Fig. 2.14 Phase diagrams for the moderately-driven pendulum, $g = 0.5$: (a) three dimensions, and (b) two dimensions ($q = 2$).

and describes a time-independent flow in phase space, similar to a set of stream lines in a fluid. In fact, the vector $\mathbf{F}$ is called a *flow*. Autonomous systems also obey the noncrossing property described earlier. However, a projection of a higher-dimensional space onto a plane might show apparent crossings which do not represent actual intersections.

The autonomous property is sufficiently useful that it is often desirable to convert a time variable to some other variable in order to make a nonautonomous system into an autonomous system. For example, the variable ϕ is introduced in the driven pendulum equations at $d\phi/dt = \omega_D$ so that the system's dynamical variables become θ , ω , and ϕ . This is convenient since the explicit time dependence enters as a *periodic* term, $g\cos(\omega_D t)$, and therefore ϕ can be a periodic variable. Then in a three-dimensional phase space, both θ and ϕ can be given periodic boundary conditions such as $\theta \in [-\pi, \pi]$ and $\phi \in [0, 2\pi]$. In Figure 2.14(a) and (b), a moderately-driven pendulum system is illustrated in both three- and

two-dimensional phase spaces. The two-dimensional diagram is a projection of the three-dimensional diagram. These diagrams show the state of the pendulum after the initial transient effects have disappeared and after the system has evolved to a steady state. The resulting closed orbit is an *attractor* in the same sense as the point is an attractor for the dissipative, nondriven pendulum. This attractor is obviously one-dimensional and is called a *limit cycle*.

The motion of the pendulum illustrated in Figure 2.14 is a simple oscillation. As the drive amplitude increases, more complex motions occur, both periodic and chaotic. Some examples of more complex periodic motions are illustrated in Figure 2.15, for various drive amplitudes (g). The path in real space and the corresponding phase plane diagram are shown in each case. The orbits involve a superposition of oscillation and complete rotation.

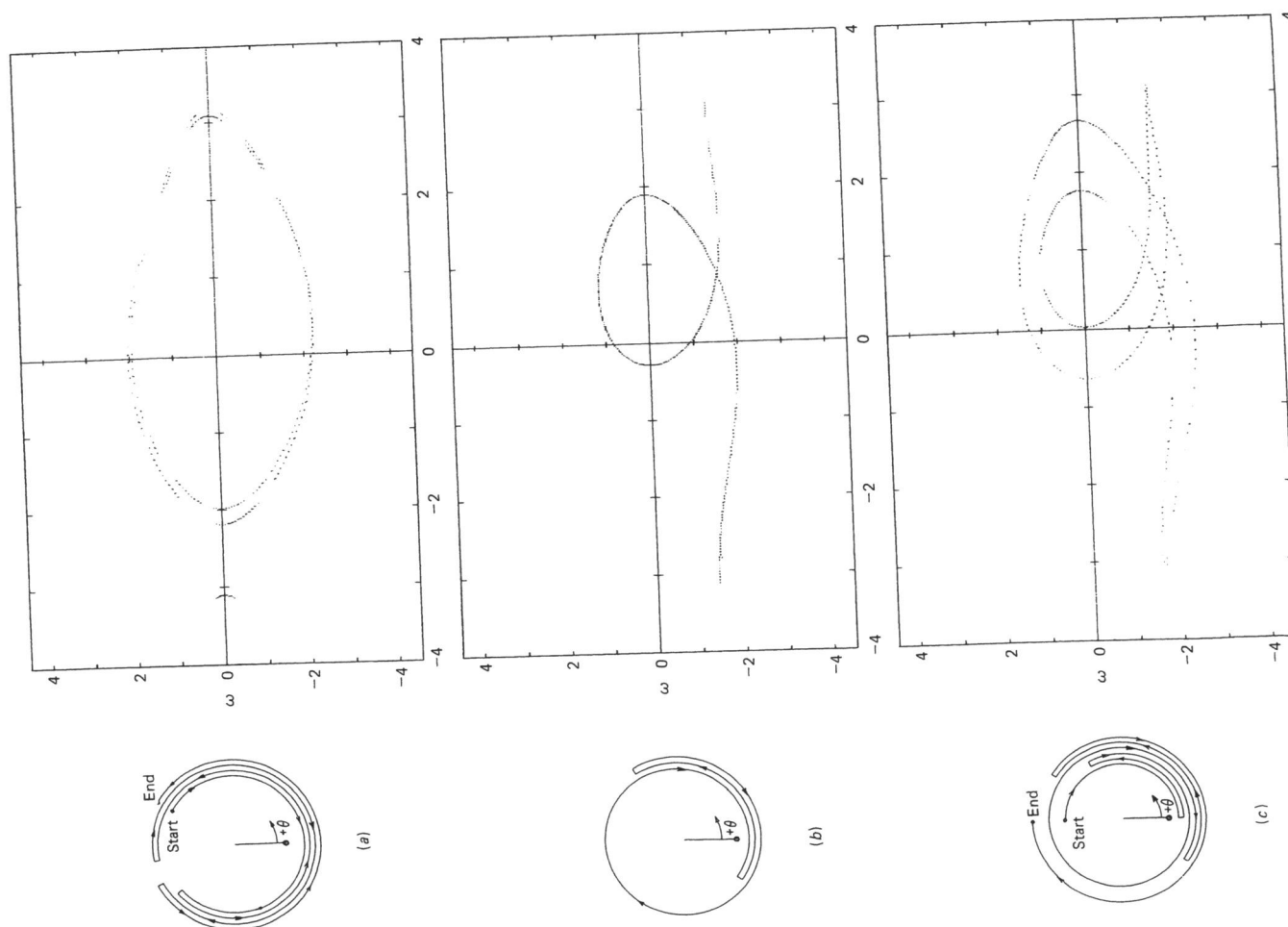
2.2 Poincaré section

A Poincaré section is a device invented by Henri Poincaré as a means of simplifying phase space diagrams of complicated systems. It is constructed by viewing the phase space diagram stroboscopically in such a way that the motion is observed periodically. For the driven pendulum, the strob period is the period of the forcing.

In order to make this idea more concrete let us refer to the moderately driven pendulum whose attractor was shown in Figure 2.14(a). The Poincaré method consists of cutting or sectioning the spiral attractor at regular intervals and looking at these sections along the ϕ axis through the (θ, ω) plane. If this sectioning is done at intervals corresponding to the forcing motion, then the stroboscopic pictures all show one point. The motion always comes back to the same (θ, ω) coordinates as ϕ is increased by 2π . Figure 2.16 illustrates the result. This strob diagram is called a *Poincaré section* as the graph is 'cut' periodically.

The Poincaré section can provide information about the ratio of the strob frequency, ω_s , to the natural frequency of the dynamical motion, ω_0 . For example, if a motion whose natural frequency was equal to 1 were strobed at a frequency equal to 2, the Poincaré section would have two points. In general, if the natural frequency of the motion, ω_0 , is equal to $(p/q)\omega_s$, where p/q is rational, then there are q points, and the order of their appearance is such that, as a given point appears on the circle, the next $[q - (p + 1)]$ positions are skipped. All of the q points, however, are

Fig. 2.15 Steady state phase diagram and sketch of the corresponding pendulum motion for different drive amplitudes. All motions are periodic and one complete cycle is shown in each case. The direction of the arrow depends on the initial conditions. (a) $g = 1.07$; (b) $g = 1.35$; (c) $g = 1.45$. ($q = 2$).



eventually filled in. Figure 2.17 illustrates the situation where, for example, when $\omega_0 = \frac{4}{5}\omega_s$, there are five points and $5 \times (4 + 1) = 0$ positions are skipped as the points go counterclockwise around the circle. These stroboscopic points provide the coordinate values for θ and ω on the Poincaré plot, and the numbering of the points represents their order of appearance.

If the pendulum goes all the way around, then ω has a direct current component as well, and the pattern of dots is not centered on the origin. Since the mixture of rotation and oscillation may lead to a nonzero average displacement $\langle \theta \rangle$, the offset will generally be asymmetric. Furthermore, if the relation between the stroboscopic frequency and the pendulum frequency is irrational (incommensurate), then the stroboscopic points will never repeat and the points will gradually fill in a circle on the Poincaré section. Finally, if the system becomes dissipative – in the pendulum case by the addition of a damping term – then the points on the Poincaré section will move toward the appropriate attractor. These ideas are illustrated in Figure 2.18.

For a dynamical system with a periodic forcing term, the Poincaré section provides a simplification of the phase diagram while retaining the essential features of the dynamics. Therefore it is ideally suited to the driven pendulum. In Figure 2.19, some of the periodic motions illustrated by phase diagrams in Figure 2.15 are shown as Poincaré sections. For the periodic motions, the appearance of these sections is quite simple. But in Chapter 3, where the chaotic behavior of the pendulum is described, the simplification of phase space provided by the Poincaré section is shown to be very important for an understanding of the

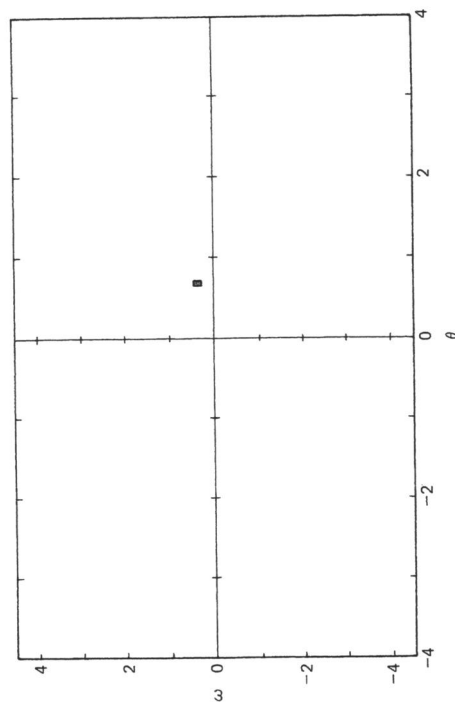


Fig. 2.16 Poincaré section of the linearized pendulum ($g = 0.5$, $q = 2$, $\phi = 0$).

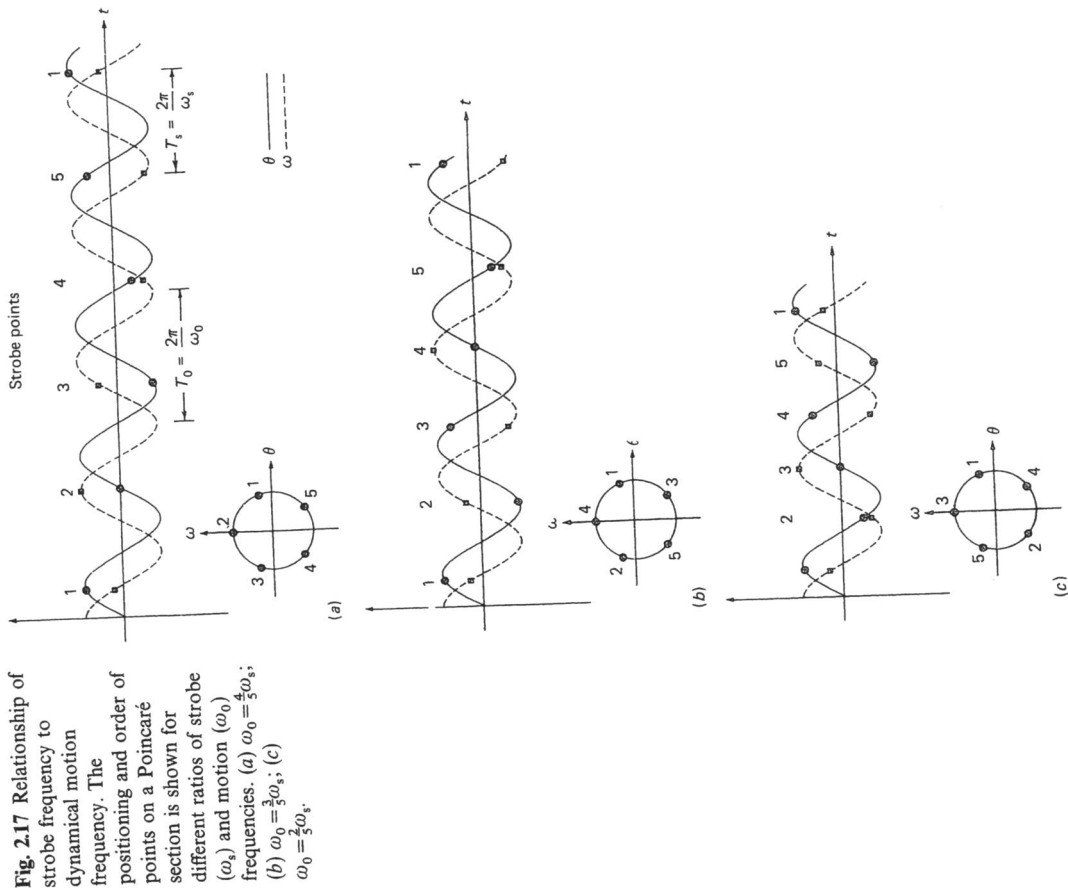
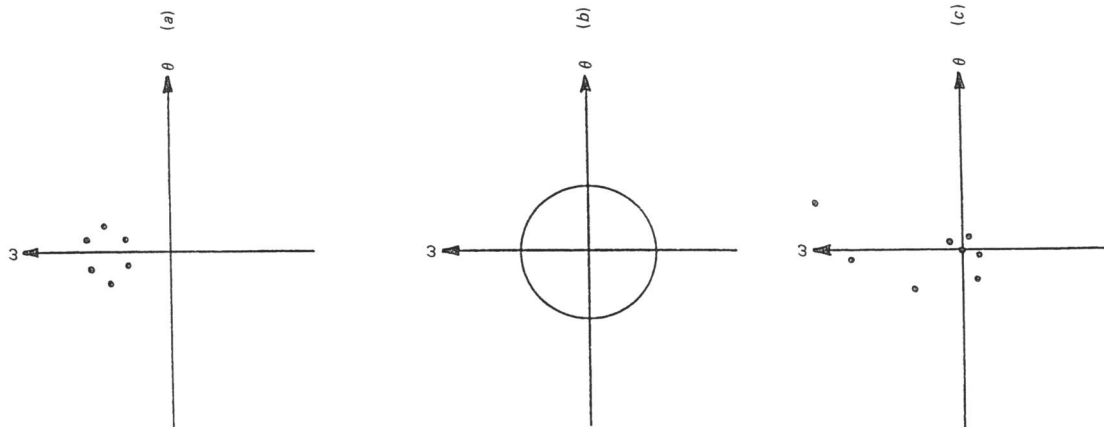


Fig. 2.17 Relationship of stroboscopic frequency to dynamical motion frequency. The positioning and order of points on a Poincaré section is shown for different ratios of strobe frequencies. (a) $\omega_0 = \frac{2}{5}\omega_s$; (b) $\omega_0 = \frac{3}{5}\omega_s$; (c) $\omega_0 = \frac{2}{5}\omega_s$.

Fig. 2.18 Poincaré sections of different motions. (a) Combined oscillatory and rotational motion whose frequency is a rational fraction of the strobe frequency. (b) Oscillatory motion whose frequency is incommensurate with the strobe frequency. (c) Dissipative motion.



physics. One useful tool for the study of chaos is the observation of the distribution of points on a computer-generated Poincaré section.

2.3 Spectral analysis of time series

The time evolution of a dynamical system is represented by the time variation $f(t)$ or (when sampled at regular intervals) *time series* of its dynamical variables. Any function $f(t)$ may be usefully represented as a superposition of periodic components. The determination of their relative strengths is called *spectral analysis*.

Depending upon the nature of the function, $f(t)$, we may represent it in two different but related ways. If $f(t)$ is periodic, then the spectrum may be expressed as a linear combination of oscillations whose frequencies are *integer multiples* of a basic frequency. This linear combination is called a *Fourier series*. However, it is more likely that $f(t)$ is not periodic, and the spectrum must then be expressed in terms of oscillations with a continuum of frequencies. Such a spectral representation is called the *Fourier transform* of $f(t)$. This representation is especially useful for chaotic dynamics. Because the Fourier transform is in general a complex-valued function, it is often preferable to define a real-valued function which is the modulus squared of the transform. This real function is called the *power spectrum* of $f(t)$. One familiar but crude example of the power spectrum is the LED display of an electronic graphics equalizer. The moving bars on the display indicate the instantaneous electronic power in each of the sections of the audio frequency spectrum.

In this section we review the main features of the Fourier series and then give the Fourier transform method as the limiting case of the Fourier series when the periodicity of $f(t)$ becomes infinitely large, that is, when $f(t)$ ceases to be periodic.

If the function is periodic such that $f(t) = f(t + nT)$ – with n being a positive or negative integer and T being the basic periodicity – then, as noted above, the frequencies of the various spectral components are all integer multiples of the basic frequency, $1/T = \omega_0/2\pi$. The *Fourier series* representation of $f(t)$ may be written compactly in complex notation:

$$f(t) = \sum_{n=-\infty}^{\infty} a_n e^{in\omega_0 t}, \quad (2.17)$$

where the a_n are the amplitudes of the components of frequency $n\omega_0$.

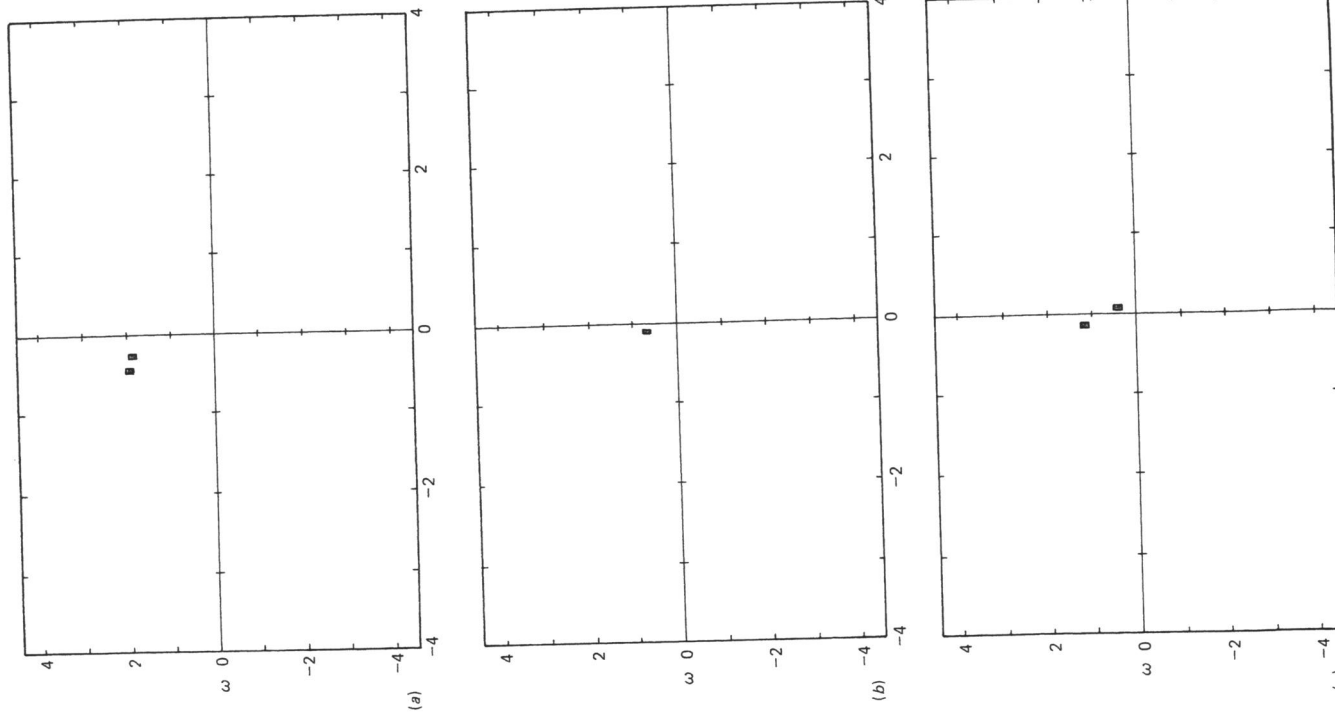


Fig. 2.19 Poincaré sections of motion illustrated in Figure 2.15. The section is taken at $\phi = 0$ ($q = 2$).
(a) $g = 1.07$; (b) $g = 1.35$;
(c) $g = 1.45$.

These amplitudes may be determined from the calculation:

$$a_n = \frac{\omega_0}{2\pi} \int_{-\pi/\omega_0}^{\pi/\omega_0} f(t) e^{-in\omega_0 t} dt. \quad (2.18)$$

(See, for example, Kaplan (1973). An example of a periodic time series often found in electronics is the 'sawtooth' function, shown in Figure 2.20.

Example 2.4

The time series of the 'sawtooth' function is $f(t) = t : t \in (-T/2, T/2)$ where $T = 2\pi/\omega_0$. (The pattern is repeated.)

It is left as an exercise to show that

$$\begin{aligned} a_n &= 0 \text{ for } n = 0; \\ a_n &= 1/in\omega_0 \text{ for } n = \text{odd integer}; \text{ and} \\ a_n &= -1/in\omega_0 \text{ for } n = \text{even integer}. \end{aligned}$$

Substitution of these results back into the Fourier series expression and manipulation of the ensuing complex expressions leads to the result:

$$f(t) = \frac{2}{\omega_0} [\sin(\omega_0 t) - \frac{1}{2}\sin(2\omega_0 t) + \frac{1}{3}\sin(3\omega_0 t) - \dots].$$

Note that the coefficients $1, -\frac{1}{2}, \frac{1}{3},$ and so forth, are not the same as the a_n but are combinations of pairs of a_n . Figure 2.20(a) shows a bar chart of the coefficients. The original function $f(t)$ and the resultant of the first two frequency components are shown as an approximation to $f(t)$ in Figure 2.20(b).

The *Fourier transform* is an extension of the Fourier series in that the basic periodicity T of $f(t)$ is allowed to become infinitely large. This condition implies that $f(t)$ need no longer be periodic. In this circumstance the spacing between the frequency components becomes infinitesimal. The discrete spectrum of frequency components becomes a *continuum* of spectral densities as shown. Therefore, a given component a_n converts to $a(\omega)\delta\omega$ where $\delta\omega$ is a small interval of frequency and $a(\omega)$ is the frequency-dependent amplitude or *Fourier transform*. The practical advantage of the transform is that it can be used to analyze a function about whose properties we are totally ignorant. It often yields surprising and illuminating information.

One may think of the transition from the Fourier series to the Fourier

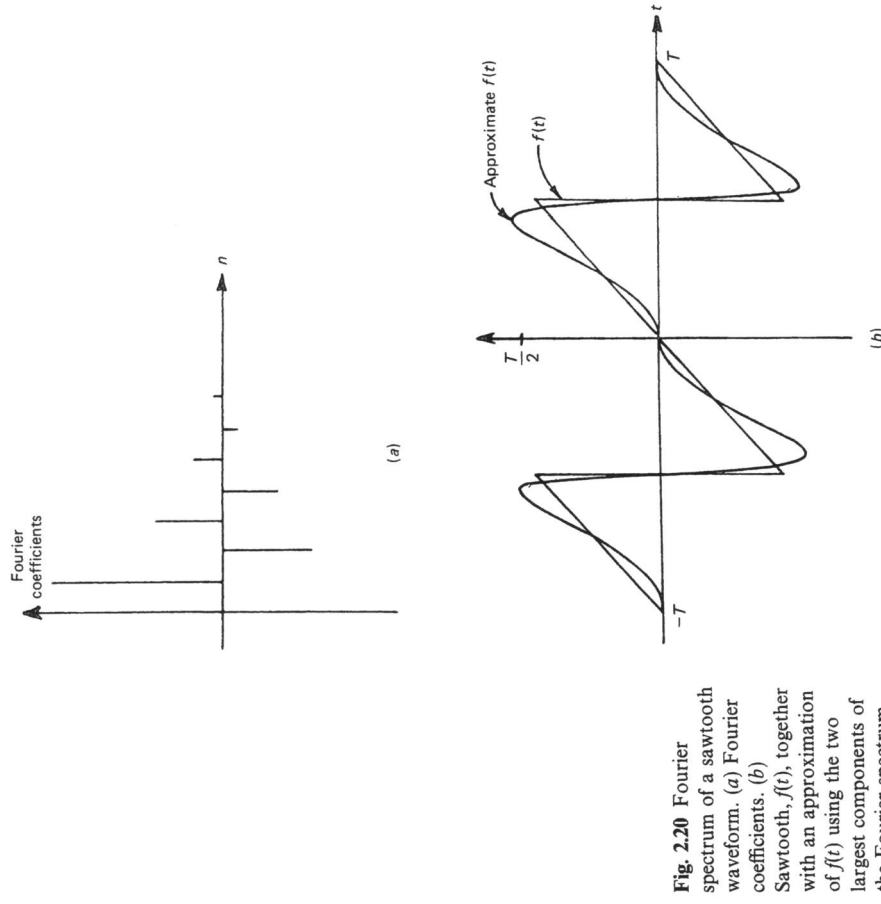


Fig. 2.20 Fourier spectrum of a sawtooth waveform. (a) Fourier coefficients. (b) Sawtooth, $f(t)$, together with an approximation of $f(t)$ using the two largest components of the Fourier spectrum.

transform in terms of the following set of transformations:

$$T \rightarrow \infty$$

$$n\omega_0 \rightarrow \omega,$$

ω being a continuous variable, and

$$a_n \rightarrow a(\omega)d\omega.$$

Taking the appropriate limits leads to the following conversions:

$$f(t) = \sum_{n=-\infty}^{\infty} a_n e^{in\omega_0 t} \text{ becomes } f(t) = \int_{-\infty}^{\infty} a(\omega) d\omega e^{i\omega t} \quad (2.19)$$

and

$$a_n = \frac{\omega_0}{2\pi} \int_{-\infty}^{\infty} f(t) e^{-in\omega_0 t} dt \text{ becomes } a(\omega) d\omega = \frac{d\omega}{2\pi} \int_{-\infty}^{\infty} f(t) e^{-i\omega t} dt. \quad (2.20)$$

The two right hand expressions lead to reciprocal expressions for the Fourier transform,

$$a(\omega) = \frac{1}{2\pi} \int_{-\infty}^{\infty} f(t) e^{-i\omega t} dt, \quad (2.21)$$

and the original function,

$$f(t) = \int_{-\infty}^{\infty} a(\omega) e^{i\omega t} d\omega. \quad (2.22)$$

As noted earlier, the Fourier transform $a(\omega)$ often turns out to be complex, and it is useful to define a real-valued function, the *power spectrum*, as

$$S(\omega) = |a(\omega)|^2. \quad (2.23)$$

One might compare this definition with the relation between wave amplitude and wave energy. The power spectrum is the quantity typically calculated in experimental or numerical work.

Let us consider two examples that can be solved analytically.

Let $f(t)$ be a decaying, oscillating function

$$f(t) = \begin{cases} e^{-\gamma t} e^{i\omega_0 t}, & t \in [0, \infty) \\ 0, & t \in (-\infty, 0] \end{cases}$$

as shown in Figure 2.21(a). The function has a natural frequency ω_0 . It could represent a dissipative, tuned electrical circuit, for example. Calculation of the integral for $a(\omega)$ leads to

$$a(\omega) = \frac{1}{2\pi[\gamma + i(\omega - \omega_0)]}$$

and then

$$S(\omega) = \frac{1}{4\pi^2[\gamma^2 + (\omega - \omega_0)^2]}.$$

This function is called a *Lorentzian* and is shown in Figure 2.21(b). $S(\omega)$ is symmetric about the dominant 'natural' frequency, ω_0 , but because of the damping parameter γ , it has a finite width.

If $\gamma \rightarrow 0$, then $S(\omega)$ becomes very sharp (approaching a delta function) with all the power then concentrated in the ω_0 component.

There is generally a reciprocal relationship between the width of $f(t)$ and the width of its Fourier transform (or $S(\omega)$). In Example 2.5, the width of $f(t)$ associated with the decaying exponential envelope is $\Delta t = 1/\gamma$, whereas the width of the Lorentzian $a(\omega)$ is $\Delta\omega = \gamma$. The product of the complementary widths is a constant of order unity. As $f(t)$ sharpens in the time domain, $a(\omega)$ broadens in the frequency domain, and vice versa.

Example 2.6

Let $f(t)$ be a wave modulated by a *Gaussian* curve:

$$f(t) = e^{-t^2/2\sigma^2} e^{i\omega_0 t}, \quad t \in (-\infty, +\infty),$$

as illustrated in Figure 2.22(a). Calculation of the integral $a(\omega)$

leads to

$$S(\omega) = e^{-(\omega - \omega_0)^2 \sigma^2},$$

which is Gaussian as shown in Figure 2.22(b). It may be shown (Problem 2.13) that the width of $f(t)$ is about $\Delta t = \sigma$, and the width of $S(\omega)$ is about $\Delta\omega = 1/(\sqrt{2}\sigma)$, the same reciprocal relationship between the widths holding as in Example 2.5.

Example 2.7

Let $f(t)$ be a linear superposition of two oscillations, with frequencies of $f_1 = 2$ and $f_2 = 3$, with amplitudes of $A_1 = 2$ and $A_2 = 1$, respectively, as illustrated in Figure 2.23(a). The corresponding power spectrum shown in Figure 2.23(b) was obtained numerically, and one easily observes the 4:1 ratio of the spectral intensities.

Example 2.8

Various types of electrical noise are easily represented by their power spectra. For example, Johnson noise which results from thermal agitation in electric circuits is frequency-independent or 'white' (in analogy with white light). On the other hand, '1/f' noise, which is common in resistors and solid state devices, has a spectrum varying as f^{-1} (or another power) at low frequencies (Malmstadt, Enke, and Crouch, 1981). These two types are illustrated in Figure 2.24.

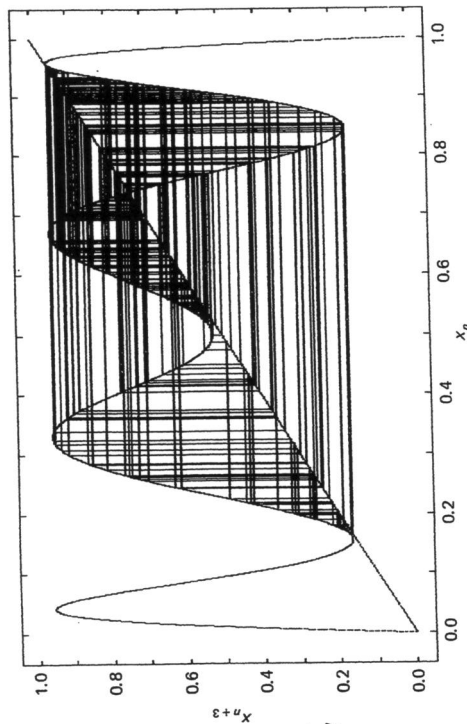


Fig. 4.8 An illustration of 'type I' intermittency as the trajectory squeezes through the gap between the map and the tangent line. During the passage through the gap, x changes very slowly.

shown in Figure 4.8. While it is in the gap, x_n is nearly fixed. If we think of the map as a Poincaré section for a differential equation representing a physical system such as the pendulum, the physical variable would show nearly periodic motion with occasionally irregular bursts. This common type of chaotic motion is called *type I intermittency* and occurs when a dynamical system is close to a tangent bifurcation if there is a mechanism for intermittent return to the narrow gap.

4.1.3 Lyapunov exponents

The Lyapunov exponent of a map (named after A.M. Lyapunov, 1857–1918, a Russian mathematician) may be used to obtain a measure of the sensitive dependence upon initial conditions that is characteristic of chaotic behavior. This exponent (often written as λ) may be readily computed for a one-dimensional map such as the logistic map. If a system is allowed to evolve from two slightly differing initial states, x and $x + \varepsilon$, then after n iterations their divergence may be characterized approximately as

$$\varepsilon(n) \approx \varepsilon e^{\lambda n}, \quad (4.9)$$

where the Lyapunov exponent λ gives the average rate of divergence. (The average must be taken over many 'initial conditions' spread over

4.1 The logistic map

the trajectory.) If λ is negative, slightly separated trajectories converge and the evolution is not chaotic. If λ is positive, nearby trajectories diverge; the evolution is sensitive to initial conditions and therefore chaotic.

Consider a specific one-dimensional map given by $x_{n+1} = f(x_n)$. The difference between two initially nearby states after the n th step is written as

$$f^n(x + \varepsilon) - f^n(x) \approx \varepsilon e^{n\lambda}, \quad (4.10)$$

or

$$\log_e \left[\frac{f^n(x + \varepsilon) - f^n(x)}{\varepsilon} \right] \approx n\lambda. \quad (4.11)$$

For small ε , this expression becomes

$$\lambda \approx \frac{1}{n} \log_e \left| \frac{df^n}{dx} \right|. \quad (4.12)$$

Finally, we use the chain rule for the derivative of the n th iterate and take the limit as n tends to infinity to obtain

$$\lambda = \lim_{n \rightarrow \infty} \frac{1}{n} \sum_{i=0}^{n-1} \log_e |f'(x_i)|. \quad (4.13)$$

Therefore the Lyapunov exponent gives the stretching rate per iteration, averaged over the trajectory. In Figure 4.9, the Lyapunov exponent is plotted as a function of the parameter μ . The sign of λ correlates very well

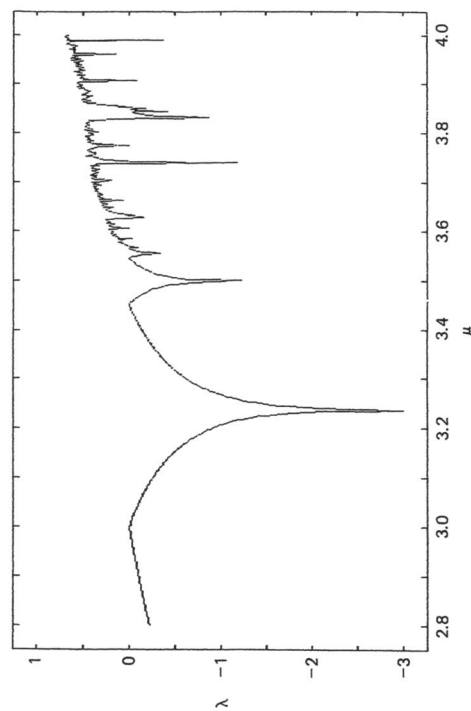


Fig. 4.9 Lyapunov exponent λ versus μ for the logistic map. Sensitivity to initial conditions occurs where the exponent is positive.

with the behavior of the system as shown in the bifurcation diagram, Figure 4.4. Beyond $\mu_\infty = 3.56$, the regions of periodic behavior correspond to the intervals in which $\lambda < 0$.

For n -dimensional maps there are n Lyapunov exponents, since stretching can occur for each axis. An n -dimensional initial volume develops, on average, as

$$V = V_0 e^{(\lambda_1 + \lambda_2 + \dots + \lambda_n)t}. \quad (4.14)$$

For a dissipative system the sum of the exponents must be negative. If the system is chaotic then at least one of the exponents is positive. (See Problem 4.14 for a soluble two-dimensional map.)

Lyapunov exponents are also defined for continuous time dynamical systems such as the pendulum. An initial n -dimensional volume of phase space develops on average as

$$V = V_0 e^{(\lambda_1 + \lambda_2 + \dots + \lambda_n)t}. \quad (4.15)$$

For the driven pendulum there are three Lyapunov exponents corresponding to the three dimensions of the phase space (θ, ω, ϕ) . Since the orbits are solutions to a set of ordinary differential equations, the calculation of Lyapunov exponents is less straightforward than for maps. On a chaotic attractor such as that of the pendulum at $g = 1.5$, the directions of divergence and contraction are locally defined, and the calculation must constantly adjust for this condition. Despite this difficulty, computer algorithms have been developed for calculating Lyapunov exponents both from differential equations and from experimental data; the Lyapunov exponents of the pendulum are discussed further in Chapter 5.

4.1.4 Entropy

The complex appearance of the various graphical representations of chaotic behavior naturally leads to the question of the relationship between statistical mechanics and chaos. One way to connect these phenomena is to apply the concept of *entropy* to a chaotic system, comparing the result to an associated statistical system. This comparison is readily done with the logistic map.

Consider a hypothetical statistical system for which the outcome of a certain measurement must be located on the unit interval. If the line is subdivided into N subintervals, we can associate a probability p_i with the i th subinterval containing a particular range of possible outcomes. The entropy of the system is then defined as

4.1 The logistic map

$$S = - \sum_{i=1}^N p_i \log_e p_i. \quad (4.16)$$

This quantity may be interpreted as a measure of the amount of disorder in the system or as the information necessary to specify the state of the system. If the subintervals are equally probable so that $p_i = 1/N$ for all i , then the entropy reduces to $S = \log_e N$, which can be shown to be its maximum value. (See Problem 4.5.) Conversely, if the outcome is known to be in a particular subinterval, then $S = 0$, the minimum value. When $S = \log_e N$, the amount of further information needed to specify the result of a measurement is at a maximum. On the other hand, when $S = 0$ no further information is required. (See, for example, Baierlein (1971) for a discussion of the interpretation of entropy as 'missing' information.)

We now apply this formulation to the logistic map by establishing N 'bins' or subintervals of the unit interval into which the values of x_n may fall. In the nonchaotic state the x_n will fall in relatively few of the bins, and the entropy is low. But in the chaotic state, the entropy is higher, and if the frequencies of occurrence are equal, then it approaches $\log_e N$. Figure 4.10 shows the results of applying the entropy concept to the logistic map. As expected, the entropy generally increases with μ , except for the downward spikes in the windows of periodic behavior. The entropy does not quite get to $\log_e N$ until $\mu = 4$, since the distribution of x_n does not span the whole unit interval evenly for $\mu < 4$. This feature may be observed in the bifurcation diagram. For $\mu = 4$, the entropy is similar to that of a random process with a uniform probability

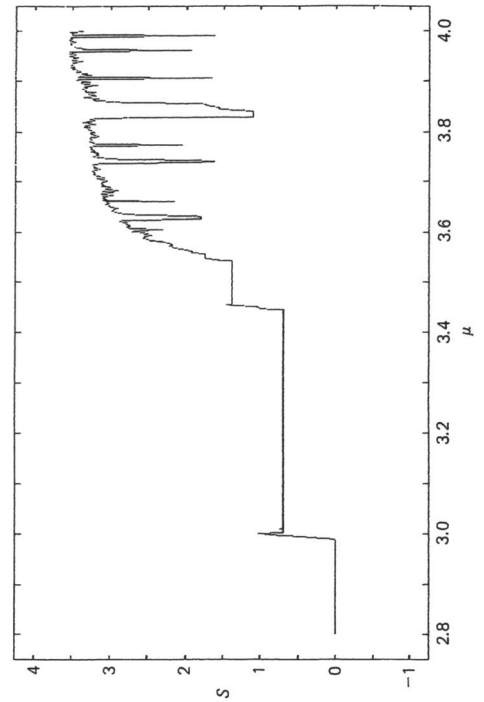


Fig. 4.10 Entropy S as a function of μ for the logistic map. The maximum entropy corresponding to equal probability for each of the 40 cells is 3.5888.

- 4.14 The baker's transformation is two-dimensional and therefore has two Lyapunov exponents, λ_x and λ_y . Determine these exponents by considering the stretching process in the x direction and the contraction in the y direction. (Answer: $\lambda_x = \log 2$, $\lambda_y = \log a$)
- 4.15 Another map that can produce horseshoes is the Hénon map:

$$\begin{aligned}x_{n+1} &= 1 - ax_n^2 + y_n, \\y_{n+1} &= bx_n.\end{aligned}$$

If $b = 1$ then the map preserves areas, and if $|b| < 1$ then the map is dissipative. Write a program to transform an initial block of (x, y) coordinates according to the Hénon map.

- 4.16 For $g > 1.4$, the pendulum exhibits period doubling. Use the BIFURCATION program through a range of g that covers the doubling region and check the ratio:

$$\frac{g_{n+1} - g_n}{g_{n+2} - g_{n+1}},$$

where g_n is the value of g at the n th bifurcation. Does this ratio appear to approach the Feigenbaum ratio?

5 The characterization of chaotic attractors

Many of the geometric structures generated by chaotic maps or differential dynamical systems are extremely complex. For example, the chaotic attractor of the pendulum (Figure 3.3(d)) in its three-dimensional phase space typically consists of an infinite number of infinitely thin layers. Its Poincaré section (Figure 3.5) reveals this structure clearly. These sets are called *fractals*. In this chapter we discuss fractals, their dimensions, and the relation between the fractal character of chaotic attractors and the underlying dynamics.

An elementary example of a fractal is the *Cantor set*. This is a prototype of complex geometric structures in much the same way that the logistic map is a prototype for chaotic dynamical systems. The Cantor set is generated by iteration of a single operation on a line of unit length, as shown in Figure 5.1. The operation consists of removing the

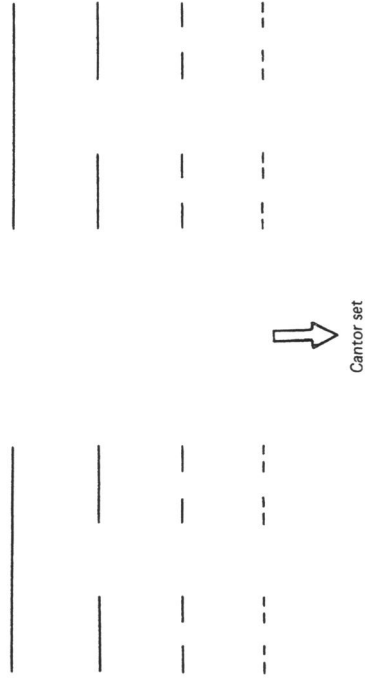


Fig. 5.1 The Cantor set, produced by the iterated process of removing the middle third from previous segments. The Cantor set has zero length and noninteger dimension.

middle third from each line segment of the previous set. As the number of iterations increases, the number of separate pieces tends to infinity, but the length of each one approaches zero. Furthermore, if the set is examined under high magnification, its structure is essentially indistinguishable from the unmagnified version. This property of invariance under a change of scale is called *self-similarity* and is common to many, although not all, fractals. (Within the resolution of the numerical simulation, the pendulum's Poincaré section of Figure 3.6 appears to exhibit this property.)

In contrast to a line with its infinite number of points and finite length, the Cantor set has an infinite number of points but zero length. Therefore it seems plausible that the dimension of the Cantor set should be less than 1 but greater than zero, the dimension of a finite set of points. The possibility of noninteger dimension requires a more sophisticated concept of dimension than that associated with lines, surfaces, and solids.

5.1 Dimension

There are many ways to define the dimension, $d(A)$, of a set A . (The particular set will often be omitted in the notation.) One approach is the *capacity* dimension, d_c .

Consider a one-dimensional figure such as a straight line or curve of length L , as shown in Figure 5.2(a). This line can be 'covered' by $N(\varepsilon)$ one-dimensional boxes of size ε on a side. If L is the length of the line then

$$N(\varepsilon) = L(1/\varepsilon). \quad (5.1)$$

Similarly, a two-dimensional square of side L can be covered by $N(\varepsilon) = L^2(1/\varepsilon)^2$ boxes as shown in Figure 5.2(b). For a three-dimensional cube the exponents would be 3, and so on for higher dimensions. In general,

$$N(\varepsilon) = L^d(1/\varepsilon)^d. \quad (5.2)$$

Taking logarithms one obtains

$$d = \frac{\log N(\varepsilon)}{\log L + \log(1/\varepsilon)}, \quad (5.3)$$

and in the limit of small ε , the term involving L becomes negligible. The *capacity* dimension is defined as

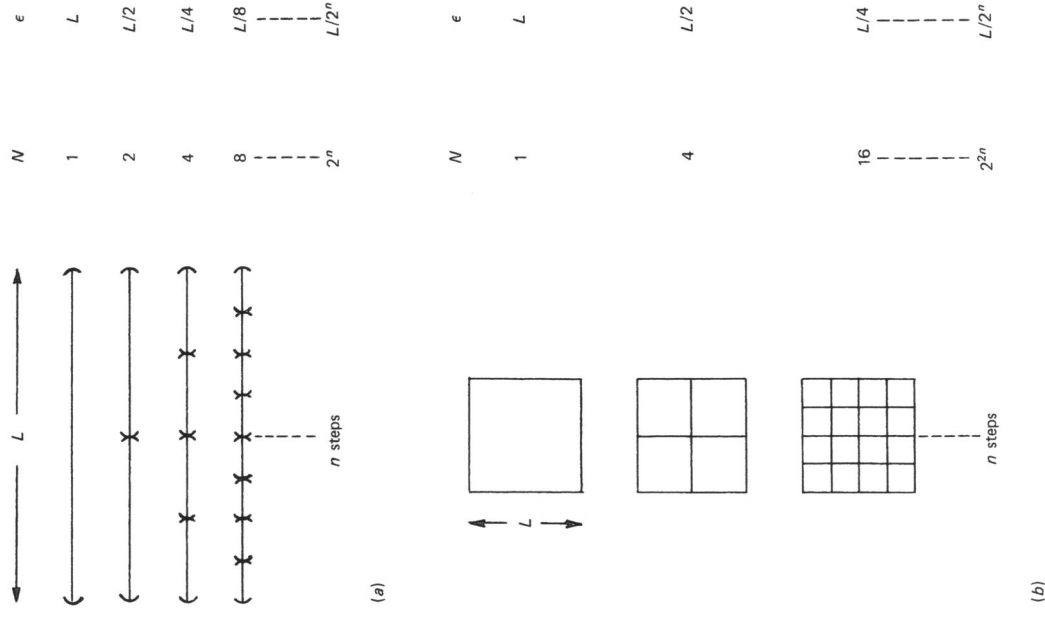


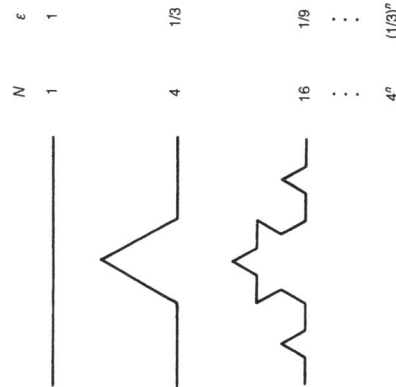
Fig. 5.2 Application of the box covering method to calculate capacity dimension. Boxes of decreasing size ε are used, leading to increasing numbers of boxes. The scaling exponent gives the dimension.

$$d_c = \lim_{\varepsilon \rightarrow 0} \frac{\log N(\varepsilon)}{\log(1/\varepsilon)} \quad (5.4)$$

An equivalent approach is to regard d_c as the slope of the $\log N$ versus $\log(1/\varepsilon)$ curve as $\varepsilon \rightarrow 0$. As shown in Figure 5.3 the Cantor set has capacity dimension $d_c = \log 2 / \log 3$.

Example 5.1

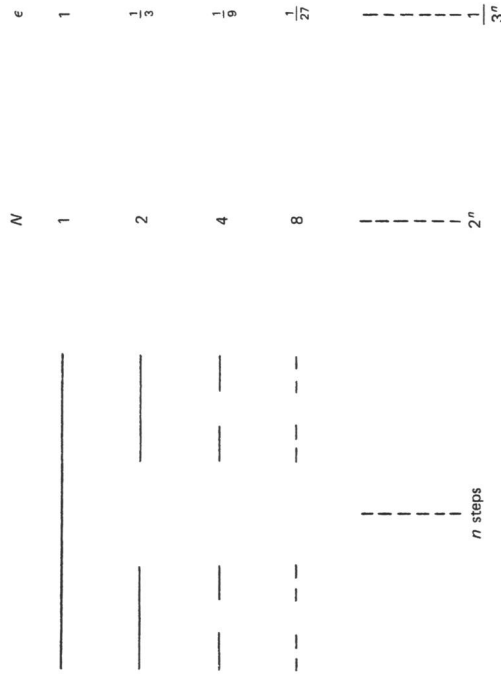
Let us calculate the dimension of the *Koch curve*. As with the Cantor set of Figure 5.3 we develop a general expression for N and ε



which leads to $d_c = \frac{\log 4}{\log 3} = 2 \frac{\log 2}{\log 3}$. (Note that this result is twice the dimension of the Cantor set.)

For a dissipative dynamical system such as the pendulum, the attractor resides in an n -dimensional phase space, but its dimension is less than n . For example, the periodic limit cycle of a lightly driven pendulum in (θ, ω, ϕ) space is an elliptical spiral with $d = 1$, as seen in Figure 3.3(a). The chaotic attractor of Figure 3.3(d), consisting of an infinite number of closely spaced sheets, has zero volume and d_c between 2 and 3. Chaotic attractors generally have noninteger dimension, and are called *strange attractors*.

Dimension calculations for dynamical systems with periodic forcing may utilize the Poincaré section rather than the full attractor (Moon and Li, 1985). The time or ϕ direction contributes 1 to the full attractor



$$\begin{aligned} d &= \lim_{\varepsilon \rightarrow 0} \left(\frac{\log N}{\log(1/\varepsilon)} \right) \\ &= \lim_{n \rightarrow \infty} \left(\frac{\log 2^n}{\log 3^n} \right) \\ &= \frac{\log 2}{\log 3} < 1 \end{aligned}$$

Fig. 5.3 Calculation of the capacity dimension for the Cantor set.

dimension, and therefore the dimension of the Poincaré section is $D_c = d_c - 1$. Hence, the Poincaré section of the lightly driven, periodic pendulum has $D_c = 0$ (it's just a point), whereas the dimension of the Poincaré section for chaotic states is between 1 and 2. Furthermore, a transverse cut across the closely spaced lines of the Poincaré section reveals a Cantor-like structure. One can think of this transverse direction as contributing the fractional part to the overall fractional dimension of the Poincaré section.

The capacity dimension is only one of several types of dimension (Farmer, Ott, and Yorke, 1983). However, any reasonable definition is expected to have the following properties:

- (a) If $A \subset B$ then $d(A) \leq d(B)$. In the case of the pendulum, the Poincaré section is a subset of the full attractor, and, in agreement with this property, the dimension of the section is less (by 1) than that of the attractor.
- (b) The null set has zero dimension.
- (c) $d(A \times B) = d(A) + d(B)$. (The symbol $\times$ indicates the Cartesian prod-

uct. If $a = \{x_i | i = 1, \dots, n\}$ and $B = \{y_j | j = 1, \dots, m\}$ then $A \times B = \{(x_i, y_j) | i = 1, \dots, n; j = 1, \dots, m\}$.) For the Cantor set in two-space, the dimension is just the sum of the dimensions of two Cantor sets in one-space:

$$\frac{\log 2}{\log 3} + \frac{\log 2}{\log 3} = \frac{\log 4}{\log 3} \quad (\text{Problem 5.3})$$

(d) If f is a differentiable and invertible function on the set A , then $d(f(A)) = d(A)$. For the pendulum the Poincaré section represents the state at a specific phase in the drive cycle. At a slightly later phase, the section is modified according to some unknown function, f , acting upon the previous state. Numerical simulation shows that the dimension remains constant, in conformity with this property.

The capacity dimension may be estimated from experimental or numerically generated data. (Implicit in such estimates is the assumption that the finite set of data points is only a sampling of an infinite set. If the data were actually limited to a finite set, then its dimension would be zero.) The technique of *box counting*, previously applied to the Cantor set, can be used for experimental or numerical data. In Figure 5.4, a portion of the pendulum's Poincaré section is shown covered with square boxes of length ε . The required number of boxes of size ε is $N(\varepsilon)$, and ε is varied to obtain $N(\varepsilon)$. From the definition given above, d_c is the slope of the graph of $\log[N(\varepsilon)]$ versus $\log[1/\varepsilon]$ as $\varepsilon \rightarrow 0$. For this choice of parameters of the pendulum $d_c = 1.3 \pm 0.1$.

For experimental data or higher-dimensional dynamical systems another type of dimension is more efficient to compute than the capacity dimension. This is the *correlation dimension*, d_G (Grassberger and Procaccia, 1983a). Suppose that many points are scattered over a set. The typical number of *neighbors* of a given point will vary more rapidly with distance from that point if the set has high dimension than otherwise. The correlation dimension may be computed from the correlation integral $C(R)$ defined by

$$C(R) = \lim_{N \rightarrow \infty} \left[\frac{1}{N^2} \sum_{i,j=1}^N H(R - |\mathbf{x}_i - \mathbf{x}_j|) \right], \quad (5.5)$$

where $\mathbf{x}_i$ and $\mathbf{x}_j$ are points on the attractor, $H(y)$ is the Heaviside function (1 if $y \geq 0$ and 0 if $y < 0$), and N is the number of points randomly chosen from the entire data set. The Heaviside function simply counts the number of points within a radius R of the point denoted by $\mathbf{x}_i$, and $C(R)$ gives the *average* fraction of points within R . Figure 5.5 illustrates the

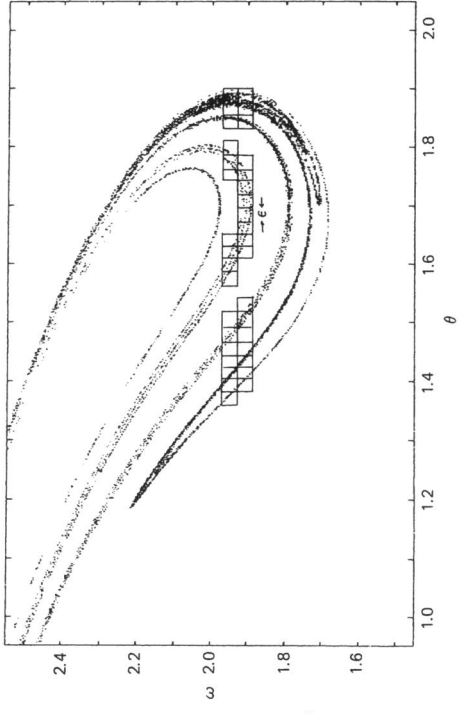


Fig. 5.4 A portion of the pendulum's Poincaré section with some representative boxes covering the points of the section. $g = 1.4954$, $q = 4$.

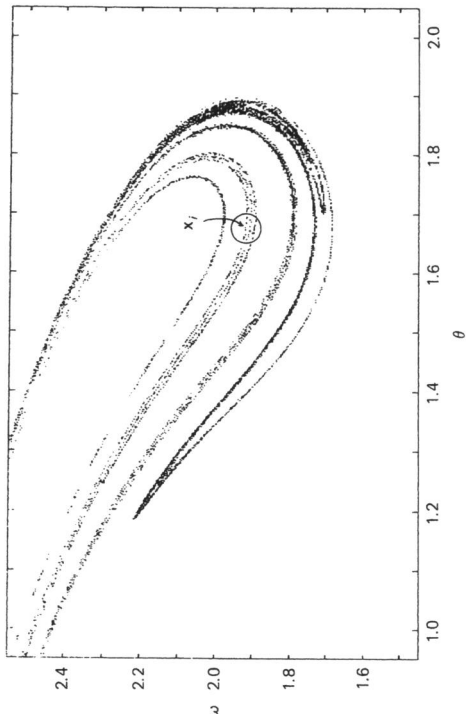


Fig. 5.5 A portion of the Poincaré section with a representative circle centered on the i th point. All the other points within the circle are counted by the Heaviside function. $g = 1.4954$, $q = 4$.

method; each point has a circle of radius R drawn around it, and then all points within all circles of size R contribute to $C(R)$. This method of estimating the dimension of a set has the advantage of using less computer memory and computational time. The correlation dimension is defined by the variation of $C(R)$ with R :

$$C(R) \sim R^{d_G} \text{ as } R \rightarrow 0. \quad (5.6)$$

Therefore the correlation dimension is the slope of a graph of $\log C(R)$ versus $\log R$. Figure 5.6 shows such a graph for the driven pendulum, and

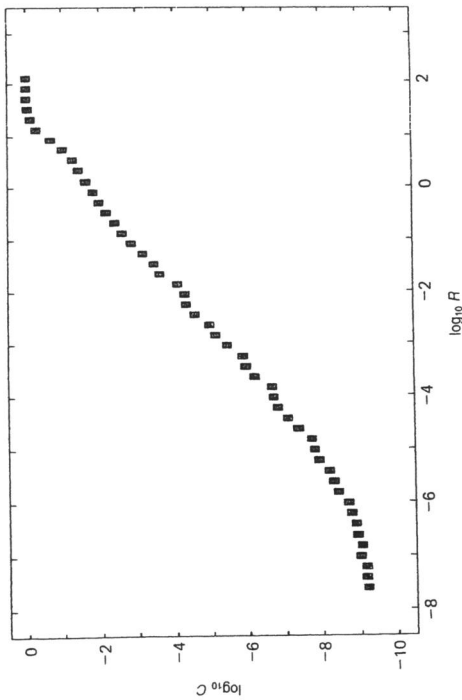


Fig. 5.6 Graph of $\log_{10} C$ versus $\log_{10} R$ for a Poincaré section of the driven pendulum. The correlation dimension is the slope of the straight line portion.

its straight line portion has a slope $d_G = 1.3 \pm 0.1$. The line differs from a straight line for high and low values of R , for the following reasons. When R approaches the size of the phase space, $C(R)$ saturates at unity since all points are then included in the circle. On the other hand, when R is smaller than the spacing between the data points, only one point lies in each circle, and $C(R)$ levels off at $1/N^2$.

The quantities d_C and d_G are not equivalent. The capacity dimension depends only on whether small elements of phase space contain *any* points and does not take into account the differing numbers of points in the various elements. That is, small scale variations of the *density* of points are ignored. On the other hand, the correlation dimension does include this effect. Because of these differences, d_C is called a *metric* dimension and d_G is called a *frequency dimension*. (See, for example, Farmer *et al.* (1983).)

A third quantity, the *information dimension* d_I , is related to the entropy defined in Chapter 4. Like d_G , the information dimension depends on the distribution of points on the attractor. Suppose the attractor is covered by a set of n boxes of size ε , and let the probability that a point is in the i th box be p_i . Then the metric entropy or missing information (see Chapter 4) is

$$I(\varepsilon) = - \sum_{i=1}^n p_i \log p_i. \quad (5.7)$$

If the points are equally distributed on the fractal, $I(\varepsilon)$ takes its

maximum value of $I_0 = \log n$. The information dimension d_I is defined by the equation:

$$d_I = \lim_{\varepsilon \rightarrow 0} \left[- \frac{I(\varepsilon)}{\log \varepsilon} \right] = \lim_{\varepsilon \rightarrow 0} \left(\frac{\sum_{i=1}^n p_i \log p_i}{\log \varepsilon} \right). \quad (5.8)$$

The information dimension is the 'scaling exponent' in the variation of the entropy, or missing information, with ε . The remarkable feature of these different definitions of dimension is that they have been shown to be related:

$$d_C \leq d_I \leq d_G. \quad (5.9)$$

If the points are uniformly distributed on the fractal, the equality holds. In practice the numerical values are usually close together (Grassberger and Procaccia, 1983a).

The above definitions of dimension have been consolidated into a general definition of dimensions of order q :

$$d^{(q)} = \frac{1}{q-1} \lim_{\varepsilon \rightarrow 0} \left(\frac{\log \sum_{i=1}^n p_i^q}{\log \varepsilon} \right), \quad (5.10)$$

where n is the number of phase space elements and p_i is the probability that an attractor point falls in the i th element. (A concise review is given in Amanspacher, Scheingraber, and Vöges 1988). It can be shown that d_C , d_I , and d_G are equivalent to $d^{(0)}$, $d^{(1)}$, and $d^{(2)}$, respectively (see Problem 5.9 and below). The higher order dimensions are sensitive to nonuniformities in the distributions of points on the attractor.

However, for a uniform distribution of points on the attractor it can be shown that all orders of generalized dimension are the same; namely, $d^{(q)} = d^{(0)}$. The derivation of this result follows directly from the fact that $p_i = 1/n$ for a uniform distribution over all n elements of the phase space for which p_i is nonzero. Then

$$\begin{aligned} d^{(q)} &= \frac{1}{q-1} \lim_{\varepsilon \rightarrow 0} \frac{\log \sum_{i=1}^n p_i^q}{\log \varepsilon} \\ &= \frac{1}{q-1} \lim_{\varepsilon \rightarrow 0} \frac{\log [n(1/n)^q]}{\log \varepsilon} \\ &= \frac{1}{q-1} \lim_{\varepsilon \rightarrow 0} \frac{\log [(1/n)^{q-1}]}{\log \varepsilon} \end{aligned}$$

$$\begin{aligned}
&= \frac{1}{q-1} \lim_{\varepsilon \rightarrow 0} \frac{(q-1) \log(1/n)}{\log \varepsilon} \\
&= \lim_{\varepsilon \rightarrow 0} \frac{\log(n)}{\log(1/\varepsilon)} = d^{(0)}.
\end{aligned} \tag{5.11}$$

But in general, $d^{(q)} \geq d^{(q')}$ where $q \leq q'$.

Let us now show the equivalence of the correlation dimension and the generalized dimension of order $q = 2$. That is, we wish to show equality of

$$d^{(2)} = \frac{1}{2-1} \lim_{\varepsilon \rightarrow 0} \frac{\log \sum_{i=1}^N p_i^2}{\log \varepsilon} \quad \text{and} \quad d_G = \lim_{R \rightarrow 0} \frac{\log[C(R)]}{\log R}.$$

The box size ε and the radius R have roughly equivalent roles, and therefore we take $R = \varepsilon$. With this assumption the proof of equivalence amounts to showing that

$$C(\varepsilon) = \frac{1}{N^2} \sum_{i,j=1}^N H(\varepsilon - |\mathbf{x}_i - \mathbf{x}_j|) \quad \text{is equal to} \quad \sum_{i=1}^n p_i^2. \tag{5.12}$$

Let us start with the correlation integral $C(\varepsilon)$. In the summations for $C(\varepsilon)$, N is both the number of circles (labeled i) and the number of points (labeled j) tested for each circle. Furthermore, the total number of circles, N , should be large enough to cover the entire attractor. In essence the circles form a covering set for the attractor. In the other summation, p_i is nonzero only for those elements of phase space which contain attractor points and therefore the upper limit of the right hand summation is also N . For these conditions

$$\begin{aligned}
\sum_{i=1}^N p_i^2 &= \sum_{i=1}^N p_i^2 \quad \text{for large enough } N. \text{ Furthermore,} \\
\sum_{i=1}^N p_i^2 &= \sum_{i=1}^N p_i p_i = \langle p_i \rangle, \quad \text{the average value of } p_i, \text{ and} \\
\langle p_i \rangle &= \frac{1}{N} \sum_{i=1}^N p_i \quad \text{also.}
\end{aligned} \tag{5.13}$$

We now connect this result to the circles of the correlation integral.

$$\begin{aligned}
\frac{1}{N} \sum_{i=1}^N p_i &= \frac{1}{N} \sum_{i=1}^n \sum_{j=1}^n (\text{probability of points in } i\text{th circle}) \\
&\approx \frac{1}{N} \sum_{i=1}^N (\text{average number of points in any circle})
\end{aligned}$$

$$= \frac{1}{N^2} \sum_{i,j}^N H(\varepsilon - |\mathbf{x}_i - \mathbf{x}_j|), \tag{5.14}$$

which is the desired result. (We emphasize that this result is rigorously true only as both N and $1/\varepsilon$ approach infinity. See Ding *et al.* (1993b) for a fuller discussion of this derivation.)

Are so many dimensions really needed to describe an attractor? The answer is yes; strange attractors, like many other fractals, contain more geometrical information than can be described by a single number, in much the same way that the mass distribution of an extended object requires an infinite number of parameters (the 'moments' of a mass distribution) for a complete characterization.

In recent studies of strange attractors, an alternative to the use of these generalized dimensions has emerged. It is based on the idea that complex sets may often be described as 'multifractals', that is, sets consisting of many interwoven fractals, each with a different fractal dimension. The multifractal concept, which is now well substantiated experimentally, leads to characterization of the set by a function called the multifractal spectrum. This function contains the same information as the generalized dimensions $d^{(q)}$, so we will not discuss it in this book. However, an accessible account is provided in Glazier and Libchaber (1988). (See also Halsey *et al.* (1986).)

While the calculation of dimension is a common feature of the research literature on strange attractors, misleading results are often obtained, as shown by Arneodo, Grasseau, and Kostelich (1987), and microcomputers are not well suited to these calculations. Therefore we approach the dimension calculation through a remarkable connection between the dimension of the attractor and the time variation of the motion, as characterized by the Lyapunov exponents. In the next two sections we develop this aspect of the subject.

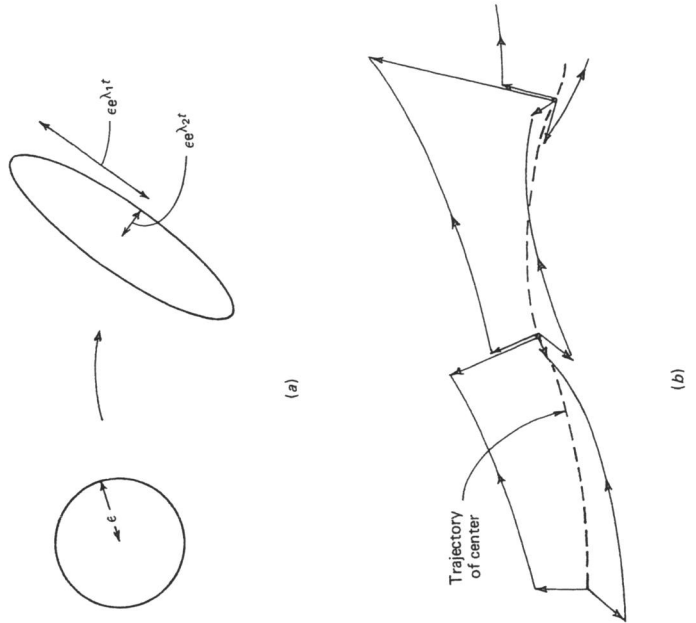
5.2 Lyapunov exponents

In Chapter 4 we introduced the idea of the Lyapunov exponent and gave an example for the one-dimensional map. Constraints on the exponents were also given for chaotic, dissipative systems in higher-dimensional spaces. For these systems, of which the pendulum is an example, the conditions are that the sum of all the exponents must be negative, but at least one must be positive to allow sensitive dependence on initial conditions.

For higher-dimensional systems, the calculation of Lyapunov exponents is more challenging than in the one-dimensional case. However, the idea is the same: measurement of the average rate of divergence of neighboring trajectories on the attractor. (See Wolf (1986) for a review article which gives an overview of the subject.)

The direction of maximum divergence or convergence is a changing local property on the attractor. The motion must be monitored at each point along the trajectory. Therefore, a small sphere is defined whose center is a given point on the attractor and whose surface consists of phase points from nearby trajectories. As the center of the sphere and its surface points evolve in time, the sphere becomes an ellipsoid, with principal axes in the directions of contraction and expansion. The evolution of the ellipsoid is illustrated in Figure 5.7(a). The average rates of expansion or contraction along the principal axes are the Lyapunov exponents. For the i th principal axis, the corresponding exponent is defined as

Fig. 5.7 A
two-dimensional example of the calculation of Lyapunov exponents. (a) The evolution of a sphere of initial point to an ellipsoid. (b) Readjustment of the size of vectors along the principal axes. Beginning at t_1 , an orthonormal set of vectors from the center of the sphere evolves by stretching and contracting along the axes of the developing ellipsoid. At t_2 , a new set of orthonormal vectors is generated such that one is parallel to the previous stretching direction. The process is repeated over many drive cycles to determine average rates of divergence and convergence of nearby trajectories.



$$\lambda_i = \lim_{t \rightarrow \infty} \left\{ \frac{1}{t} \log_e [L_i(t)/L_i(0)] \right\}, \quad (5.15)$$

where $L_i(t)$ is the radius of the ellipsoid along the i th principal axis at time t (Wolf *et al.*, 1985). In this expression, the growth rate is always measured along the i th principal axis, but the absolute orientation in phase space of that axis is not fixed. It is impractical to perform the actual computation in the way suggested by the definition, because the initially close phase points would soon diverge from each other by distances approaching the size of the attractor, and the computation would then fail to capture the local rates of divergence and contraction. Therefore, vectors connecting the surface of the ellipsoid to the center must be shrunk periodically or *renormalized* to ensure that the size of the ellipsoid remains small and that its surface points correspond to trajectories near that of the center point. The renormalization is illustrated in Figure 5.7(b). The Lyapunov exponents are taken to be the *averages* of those obtained over many segments of the central trajectory.

The algorithm for the Lyapunov exponent calculation in more than one dimension is somewhat beyond the scope of this text, although a computer program that calculates the exponents for the pendulum is included in Appendix B. The method is described in some detail in Wolf *et al.* (1985), and FORTRAN code given in that article is the basis for the program listing.

Nevertheless, it is useful to discuss some of the results for the pendulum. Figure 5.8 shows a typical graph of the three computed Lyapunov exponents for the pendulum, plotted as a function of the number of drive cycles utilized for the computation. The initial transients gradually decay as the calculation is extended over many cycles, so that the structure of the attractor is fully explored.

There are three Lyapunov exponents because the pendulum equations have three variables, and their sum should be negative since the system is dissipative. One exponent corresponds to the direction parallel to the trajectory. It contributes nothing to the expansion or contraction of phase volumes, and therefore the corresponding Lyapunov exponent (λ_2) is zero.

The remaining exponents are negative or zero in the periodic states, whereas in the chaotic state one exponent is positive, indicating divergence of trajectories. For the parameters used to generate Figure 5.8 the system is chaotic, and (from the figure) the exponents are estimated to be: $\lambda_1 = 0.16$, $\lambda_2 = 0$, and $\lambda_3 = -0.42$, with an uncertainty of about 0.02.

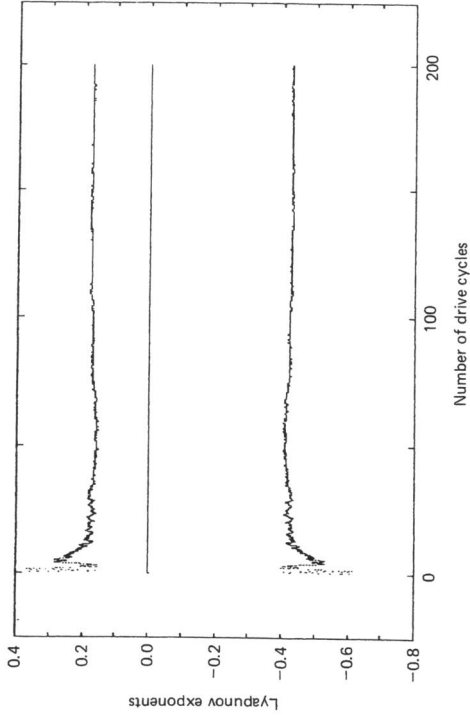


Fig. 5.8 Graph of the three Lyapunov exponents for the chaotic pendulum versus the number of orbits (or drive cycles) used in the computation. Here $g = 1.4954$ and $q = 4$. After a few cycles, the steady state values are reached.

An interesting relationship can be developed between the Lyapunov exponents and the autonomous set of differential equations describing the pendulum. This relation depends on the fact that both the Lyapunov exponents and the differential equations contain information on the change of volumes in phase space, and the relation provides a check on the numerical values developed from the Lyapunov algorithm. First we note that, according to the definition of Lyapunov exponents, a small volume in phase space will change in time as

$$V(t) = V_0 e^{(\lambda_1 + \lambda_2 + \lambda_3)t}, \quad (5.16)$$

and therefore the rate of change of $V(t)$ is simply

$$dV/dt = \sum_{i=1}^q \lambda_i V(t). \quad (5.17)$$

Furthermore, the discussion of Chapter 2 concerning the change of phase volume in a time δt showed that the time rate of change of the phase volume can also be written as

$$dV/dt = [\nabla \cdot \mathbf{F}] V, \quad (5.18)$$

where $\mathbf{F}$ is the vector function forming the right side of the autonomous set of first order differential equations for the dynamical system. In the case of the pendulum, $\nabla \cdot \mathbf{F}$ equals $-1/q$, independently of position and time. The two expressions for rate of change of volume are combined and lead to the simple result:

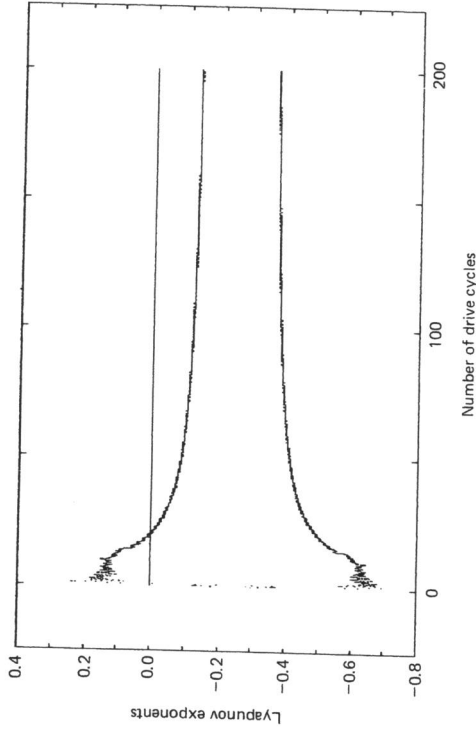


Fig. 5.9 Graphs of Lyapunov exponents computed for a periodic pendulum, with $g = 1.125$ and $q = 2$. The steady state values can be used to check the relation,

$$1/q = -\sum_{i=1}^q \lambda_i.$$

$$1/q = -\sum_{i=1}^3 \lambda_i. \quad (5.19)$$

The exponents obtained from Figure 5.8 with $q = 4$ satisfy this relation. The Lyapunov exponents for a periodic state (Figure 5.9) also satisfy the relation to within numerical precision.

5.3 Lyapunov exponents and dimension

An interesting and important relationship between Lyapunov exponents and dimension was proposed by Kaplan and Yorke (1979). This conjecture can be developed heuristically by reference to Figure 5.10 where a region in (θ, ω) phase space evolves in time under the influence of a dissipative, chaotic system. In this simple example, one direction stretches by a factor of $e^{\lambda_1 t}$ with $\lambda_1 > 0$, and the other direction shrinks according to the factor $e^{\lambda_2 t}$ with $\lambda_2 < 0$. Therefore the area evolves as $A(t) = A_0 e^{(\lambda_1 + \lambda_2)t}$. Now we define a Lyapunov dimension d_L by analogy with the capacity dimension as

$$d_L = \lim_{\varepsilon \rightarrow 0} \left[\frac{d(\log(N(\varepsilon)))}{d(\log(1/\varepsilon))} \right], \quad (5.20)$$

where $N(\varepsilon)$ is the number of squares with sides of length ε required to cover $A(t)$. Both $N(\varepsilon)$ and the length ε depend on time as follows:

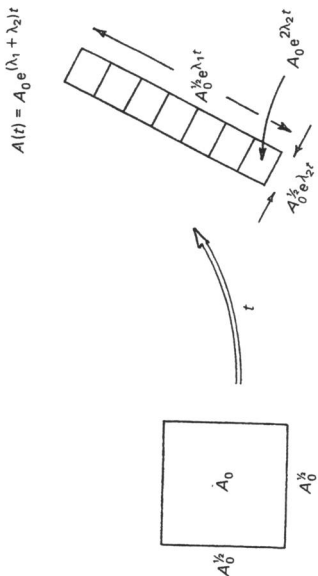


Fig. 5.10 Schematic diagram of the role of Lyapunov exponents in the stretching and contraction of an area in phase space.

$$N(t) = \frac{A(t)}{\text{square area}(t)} = \frac{A_0 e^{(\lambda_1 + \lambda_2)t}}{A_0 e^{2\lambda_2 t}} = e^{(\lambda_1 - \lambda_2)t} \quad (5.21)$$

and

$$\varepsilon(t) = A_0^{\frac{1}{2}} e^{\lambda_2 t}.$$

Combining these expressions with the definition of d_L and using the chain rule, it follows that

$$d_L = 1 - \frac{\lambda_1}{\lambda_2}. \quad (5.22)$$

This is the Kaplan-Yorke relation. Kaplan and Yorke proposed that $d_L \geq d_G$ and that if the points on the fractal are approximately uniformly distributed, then the equality should hold. (The above derivation depended on a simple example of contraction and expansion in fixed, mutually perpendicular directions. A discussion of the conditions under which the conjecture should be true has been given by Grassberger and Procaccia (1983a).)

The Kaplan-Yorke relation may be generalized to higher-dimensional spaces. The formula becomes

$$d_L = j + \frac{\lambda_1 + \lambda_2 + \lambda_3 + \dots + \lambda_j}{|\lambda_{j+1}|}, \quad (5.23)$$

where the λ_i are ordered (λ_1 being the largest) and j is the index of the smallest nonnegative Lyapunov exponent. For example, in the phase space (θ, ω, ϕ) the Lyapunov exponents for the pendulum were found to

Table 5.1. Dimensions of the Poincaré section for various damping factors. All states shown are chaotic. Decreasing the damping allows the Poincaré section to spread and the dimension to increase. ($g = 1.5$)

q	λ_1	λ_2	$\sum_i \lambda_i$	$\frac{1}{q}$	$d = 1 + \frac{\lambda_1}{ \lambda_2 }$
4.0	0.16	-0.42	-0.26	0.25	1.38
3.7	0.16	-0.43	-0.27	0.27	1.37
3.0	0.11	-0.44	-0.33	0.33	1.25
2.8	0.09	-0.45	-0.36	0.36	1.2
2.0	0.12	-0.58	-0.46	0.5	1.2

be $\lambda_1 = 0.16$, $\lambda_2 = 0$, and $\lambda_3 = -0.42$ (when $g = 1.4954$ and $q = 4$). Therefore $j = 2$ and the predicted dimension is

$$d_L = 2 + \frac{0.16 + 0}{0.42} \approx 2.4.$$

In the two-dimensional subspace (θ, ω) the exponents are $\lambda_1 = 0.16$ and $\lambda_2 = -0.42$ and the calculated dimension is $d_L = 1 + \frac{0.16}{0.42} \approx 1.4$, with an uncertainty of 0.05. (Remember that the ϕ dimension corresponds to time and does not contribute to the expansion or contraction of phase volume.) The result $d_L = 1.4$ correlates well with values of both the capacity and correlation dimension calculated directly for this particular state of the pendulum.

The Lyapunov exponents, in conjunction with the Kaplan-Yorke relation, can be used to study chaotic pendulum states with different amounts of damping. For example, when $q = 2$ the pendulum is highly damped, points in the Poincaré section are tightly packed, and the dimension is small. Table 5.1 shows Lyapunov exponents for chaotic states with various damping factors. The corresponding calculated dimensions are also given. While the changes indicated are not large, there is a general increase in dimension with decreasing damping. However, the pendulum is not chaotic for every value of q in the range shown.

From this discussion it is evident that Lyapunov exponents provide an important link between the fractal geometry of the attractor and the property of sensitive dependence on initial conditions.

5.4 Information change and Lyapunov exponents

The condition of sensitivity to initial conditions that is characteristic of chaotic systems implies divergence of initially adjacent dynamical states. While an initial state of the system may be known with a high but finite degree of precision, the ability to predict later states diminishes because of trajectory divergence. Information is lost, or conversely, more information is required to specify the system with the original precision; the entropy has increased.

For many systems the information function has a simple linear time dependence (Atmanspacher and Scheingraber, 1987)

$$I(t) = I_0 + Kt. \quad (5.24)$$

In this expression, $I(t)$ depends on the probabilities p_i , which change with time as the system evolves. Note that I also depends on the length, ε , of the elemental unit that is used to partition the space. The rate of information change or *Kolmogorov entropy* K is an average rate taken over long times T , although in a numerical calculation T would have to be less than the time taken to cover the entire attractor. Subject to these conditions, K is defined as

$$K = \lim_{\varepsilon \rightarrow 0} \lim_{T \rightarrow \infty} [I(\varepsilon, T)/T]. \quad (5.25)$$

(See, for example, Grassberger and Procaccia (1984).)

As an illustration of Kolmogorov entropy, consider an imaginary dynamical model which is capable of (a) deterministic, nonchaotic motion, (b) deterministic, chaotic motion, and (c) random motion, for which future states are completely unknown. The value of K will depend on the type of motion. We assume that the phase space of the system may be divided into N small regions, each with probability p_i . Suppose that the system is known to be in a particular initial phase region, and that all initial probabilities are zero except the corresponding probability for that region, which is one. Therefore the initial information (entropy) is zero. The three kinds of motion will cause different types of evolution and therefore different changes in the information function.

For nonchaotic time evolution the system's phase trajectories remain close together. After a time T , nearby phase points are closely grouped in some other small region of phase space and there is no change in information. Therefore the Kolmogorov entropy is zero. For chaotic

evolution phase trajectories diverge, and the number of phase space regions available to the system after a time T is

$$e^{\lambda_+ T}$$

where λ_+ is a positive Lyapunov exponent. Assuming that all of these regions are equally likely, the information function now becomes

$$I(T) = - \sum_{i=1}^N p_i(T) \log_e p_i(T) = \log_e e^{\lambda_+ T} = \lambda_+ T. \quad (5.26)$$

Therefore the Kolmogorov entropy in the chaotic case is λ_+ . Finally, for random evolution all phase space regions become possible after a very short time, and assuming equal probabilities, the information entropy is

$$I = \log N \quad (5.27)$$

where N is the number of phase space cells. Although N is finite, the fact that each cell has a finite probability of being instantly accessible implies an infinite rate of information expansion. Therefore, in the case of random evolution the Kolmogorov entropy is infinite.

From this discussion it is evident that the Kolmogorov entropy for chaotic systems depends on the positive Lyapunov exponent. In systems with several positive exponents,

$$K \leq \sum_{i=1}^j \lambda_i \quad (5.28)$$

where j is the index of the smallest positive λ_i . Grassberger and Procaccia (1983a) suggest that equality usually holds. Hence knowledge of the Lyapunov exponents provides a good estimate of Kolmogorov entropy, and positive but finite K implies chaotic behavior. Furthermore, the Kolmogorov entropy provides a means of categorizing the motion of dynamical systems (Atmanspacher and Scheingraber, 1987). If $K=0$, then the motion is regular (periodic, quasiperiodic, or stationary). If $K>0$, then the motion is chaotic, and if $K=\infty$, then the motion is random.

Another feature of the Kolmogorov entropy is that it may be used to estimate the time for which future predictions of the state of a chaotic system are valid. After this prediction time, the system's uncertainty has diverged to the size of the phase space (at least in some direction), and the folding action has to occur. Then the system can be described only by a probability density that depends on location in phase space. For purposes of prediction the system becomes probabilistic.

Suppose the system is characterized by a positive Lyapunov exponent, λ_+ , and its initial state is defined to within a size ε . Then, in a time, T , the uncertainty in the coordinates will have expanded to the size L of the attractor:

$$L \sim \varepsilon e^{\lambda_+ T} \text{ or } L \sim \varepsilon e^{KT} \quad (5.29)$$

Either of these relations may be solved for the prediction time:

$$T \sim (1/\lambda_+) \log_e(L/\varepsilon) \text{ or } T \sim (1/K) \log_e(L/\varepsilon). \quad (5.30)$$

Therefore, the prediction time increases only logarithmically with the precision of the initial measurement. For this reason, chaotic states allow only short-term prediction.

In chapter 3 we raised a question about the efficacy of very long computer simulations of chaotic systems when such simulations introduced errors that would apparently be rapidly multiplied in chaotic systems. Our derivation of a formula for prediction time allows us to calculate the effect. For example, if the phase space coordinates are in error by only 1 part in 10^8 of the attractor size and $\lambda_1 = 0.16$ then the prediction time is about 115 time units. A computer simulation that introduces errors of this magnitude will then be accurate for only this relatively short prediction time. We can illustrate the effect by providing an artificial 'error' in the form of a small change of initial conditions. In Figure 5.11 we show time series of two pendula with slightly different initial conditions. The starting values of the angular velocity differ by

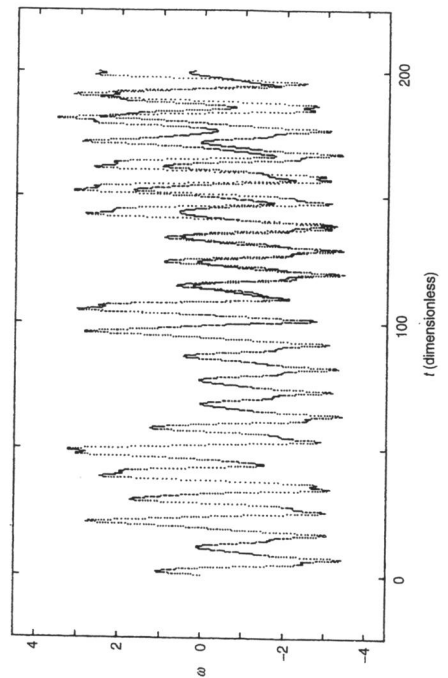


Fig. 5.11 Time series for two pendula whose initial angular velocities differ by $\Delta\omega/\omega = 10^{-8}$. The time for divergence of these systems is estimated to be 115 time units.

4×10^{-8} or one part in 10^8 relative to the attractor size. As expected, the two time series begin to diverge very close to the prediction time.

Does the divergence inherent in chaotic systems mean that strange attractors produced by long simulations are meaningless? The answer appears to be no (Grebogi *et al.*, 1990, Fryskai and Zohdy, 1992). For many chaotic systems the 'true' orbits (error-free) are very dense on the attractor in the phase space. Therefore a 'false' trajectory – one from a simulation with inherent error – is often very close to a true orbit that will shadow the false orbit. Occasionally false and true orbits will break apart and then another nearby true orbit will shadow the false orbit. The result is that, in many circumstances, an attractor developed from a computer simulation will have the same structure as an attractor for true orbits. While an individual time series from a computer simulation may not match a true orbit, the resultant attractor still has the appropriate geometry as well as the correct dimension, Lyapunov exponents, and other more sophisticated invariant measures.

Problems

- 5.1 The length of the Cantor set may be determined by subtracting the length of each segment taken out of the set during each step in its formation. For example, a length $\frac{1}{3}$ is taken out in the first step, $2(\frac{1}{3})$ in the second step, $4(\frac{1}{9})$ in the third step, and so on. Form the infinite geometric series which this process describes and show that the sum approaches 1. Therefore the Cantor set has no length.
- 5.2 Construct a fractal that is similar to the Cantor set, but instead remove the middle $\frac{1}{2}$ from each previous section. Show that its dimension is $\frac{1}{2}$.
- 5.3 Construct a 'two-dimensional' Cantor set in the following way. Draw two Cantor sets at right angles to each other so they just touch at one corner. Then fill in those two-dimensional regions where each 'one-dimensional' set intersects the other. In other words form the set which is the Cartesian product of the two original sets, as in Figure (a) below. Show that the capacity dimension is $2(\log 2 / \log 3)$. (This set resembles the invariant set of the horseshoe transformation; that is, the points which remain from the original set after many iterations.)
- 5.4 Following a procedure analogous to that used in Problem 5.1, calculate the area of the two-dimensional Cantor set defined in